

# BFS Traversal

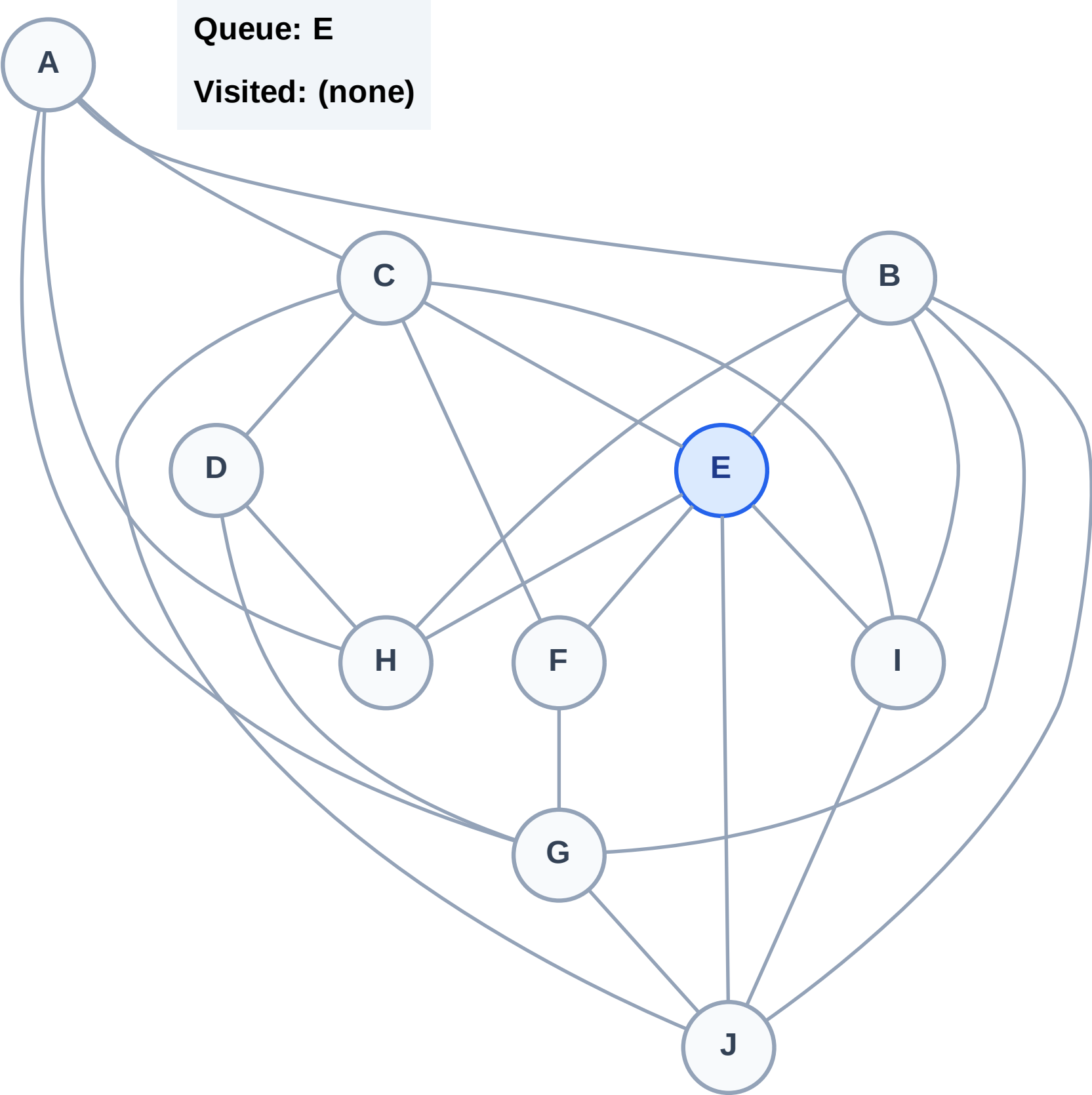
Step 1: Start at E

Complete

Processing

Discovered

Unseen



# BFS Traversal

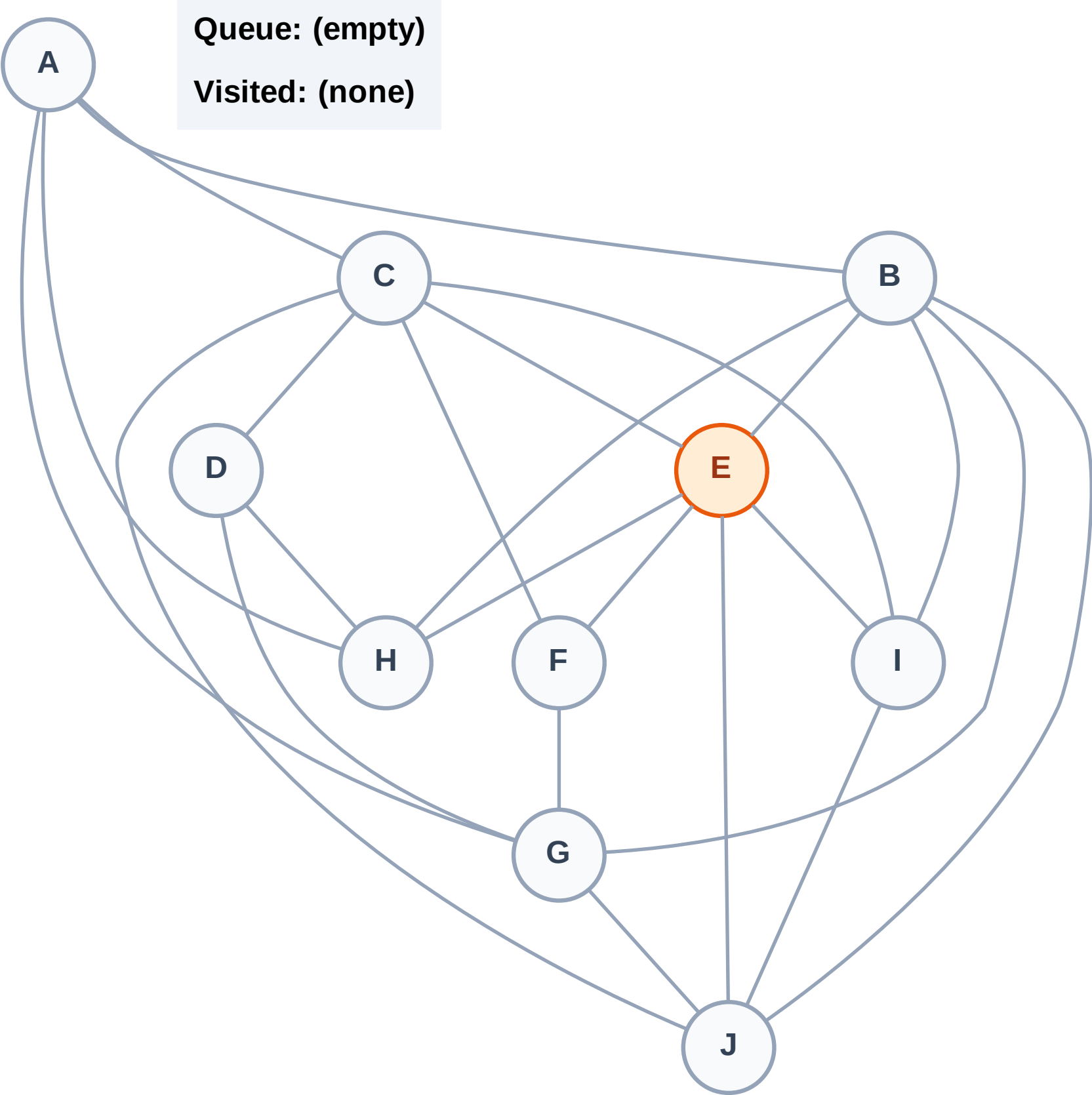
Step 2: Process node E

Complete

Processing

Discovered

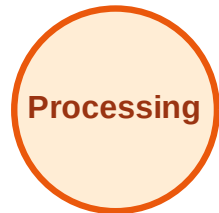
Unseen



# BFS Traversal

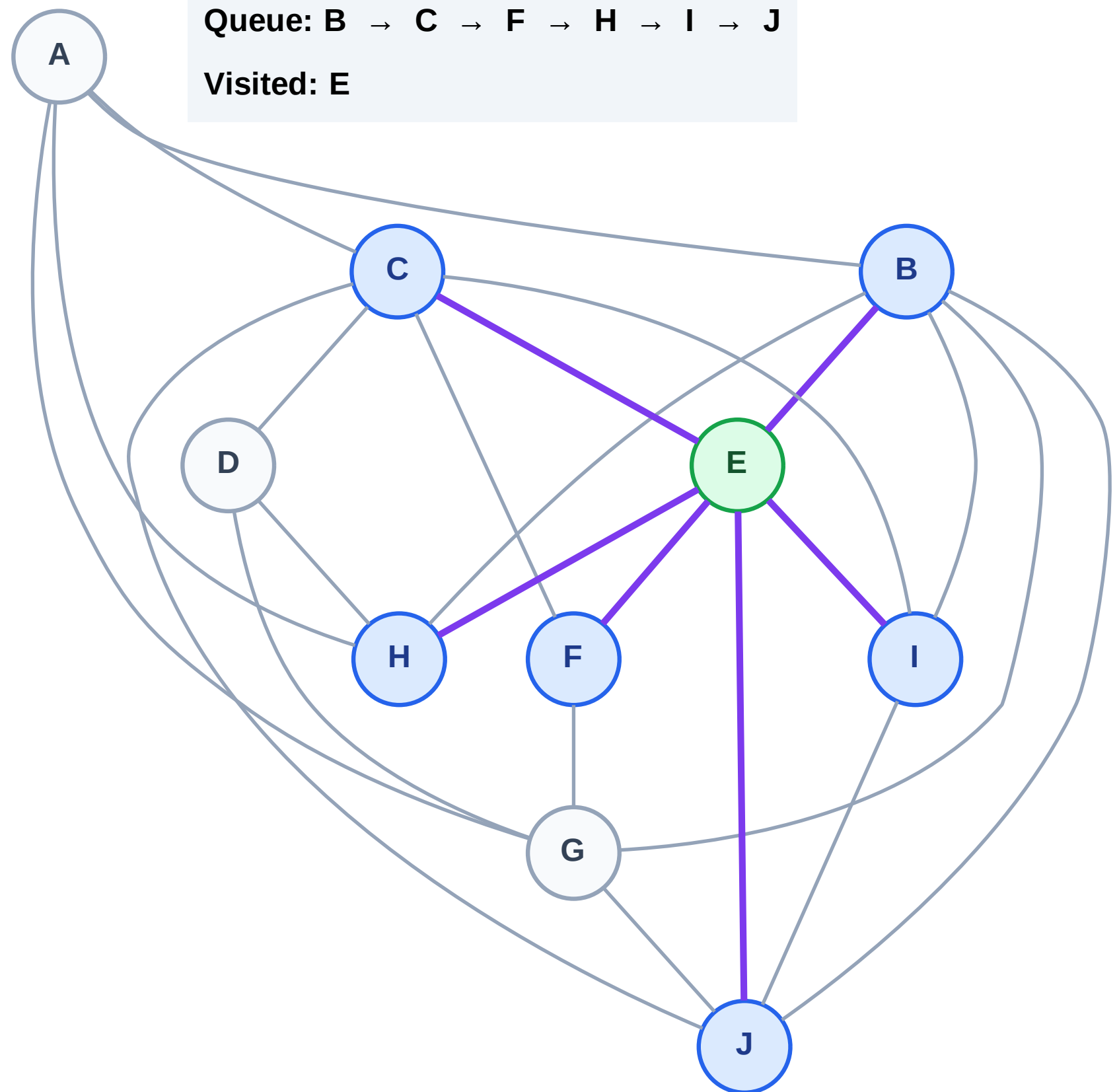
Step 3: Discover B, C, F, H, I, J from E

## Legend



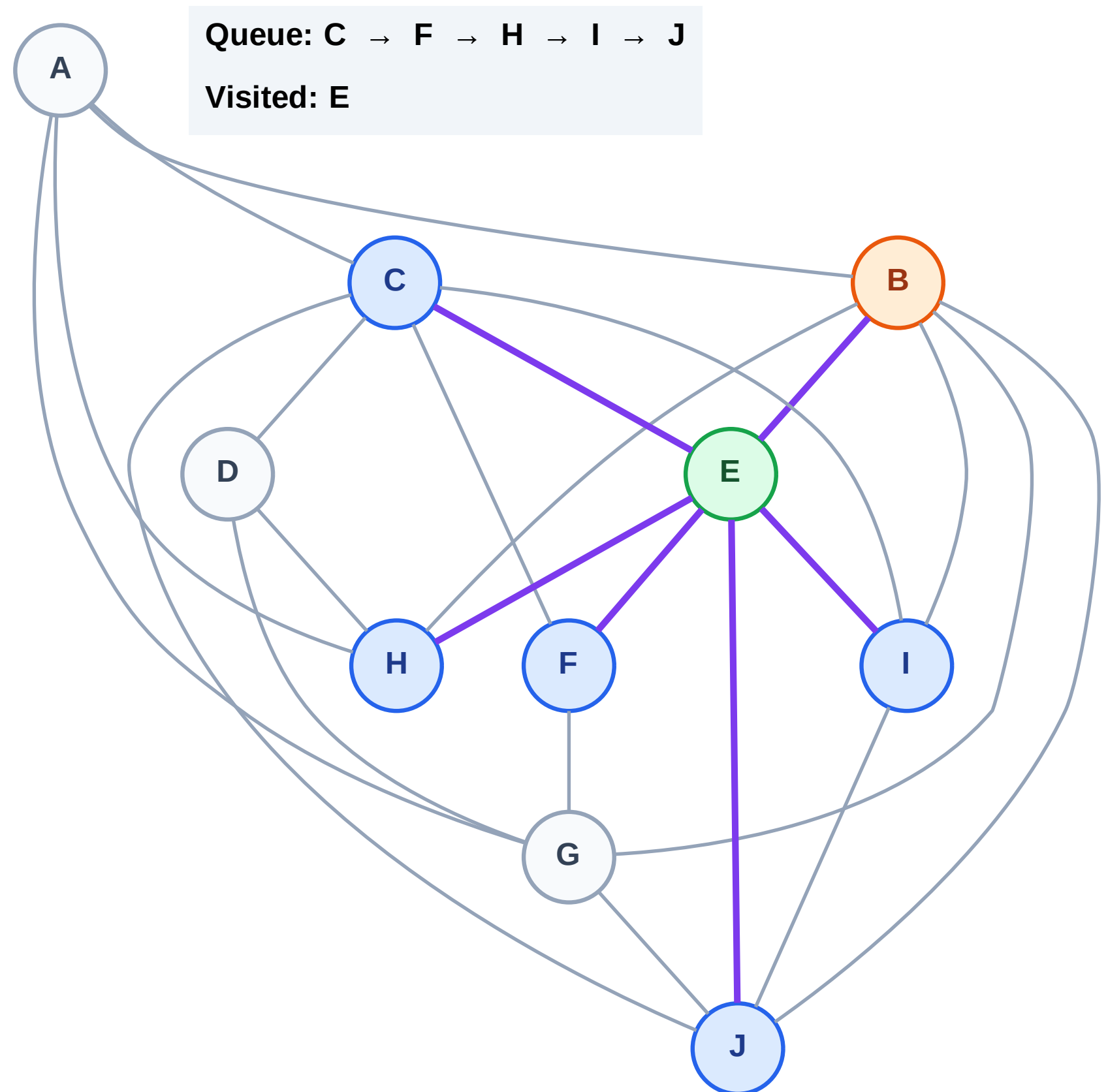
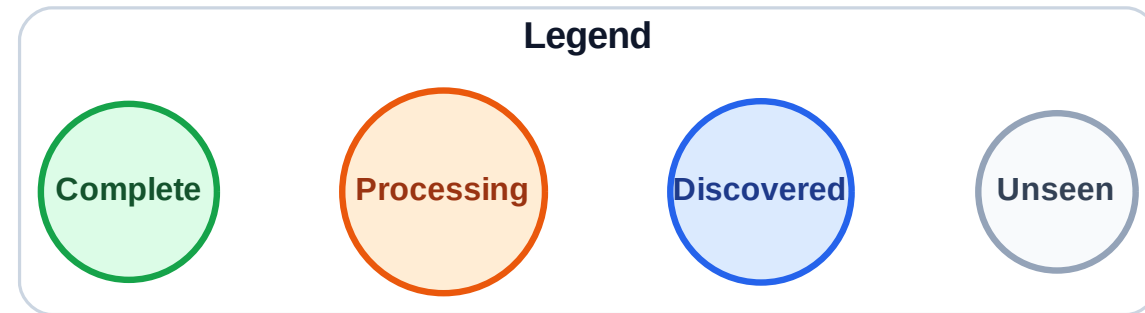
Queue: B → C → F → H → I → J

Visited: E



# BFS Traversal

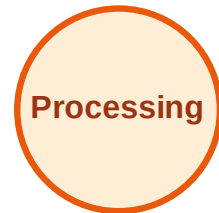
Step 4: Process node B



# BFS Traversal

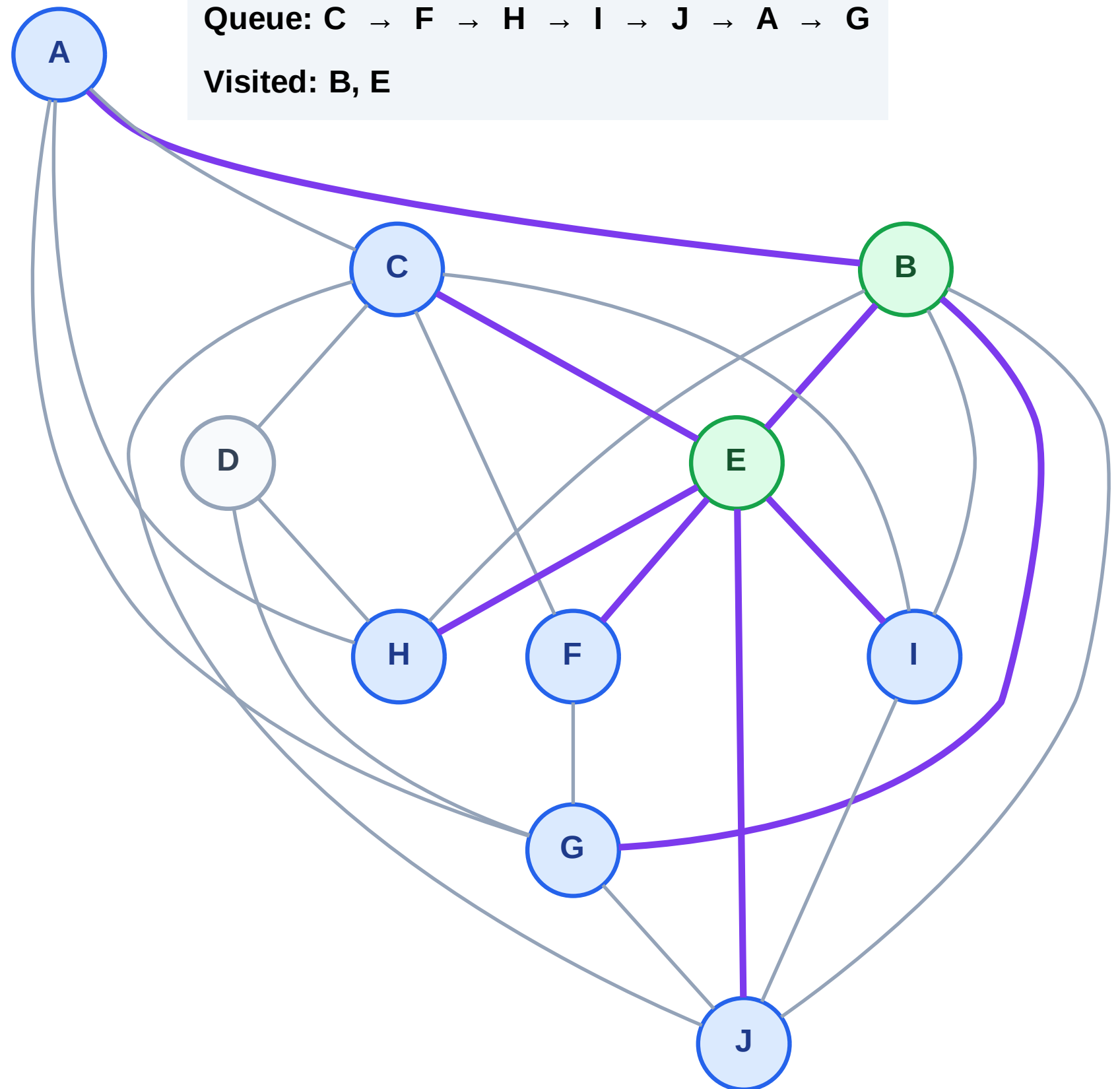
Step 5: Discover A, G from B

## Legend



Queue: C → F → H → I → J → A → G

Visited: B, E



# BFS Traversal

Step 6: Process node C

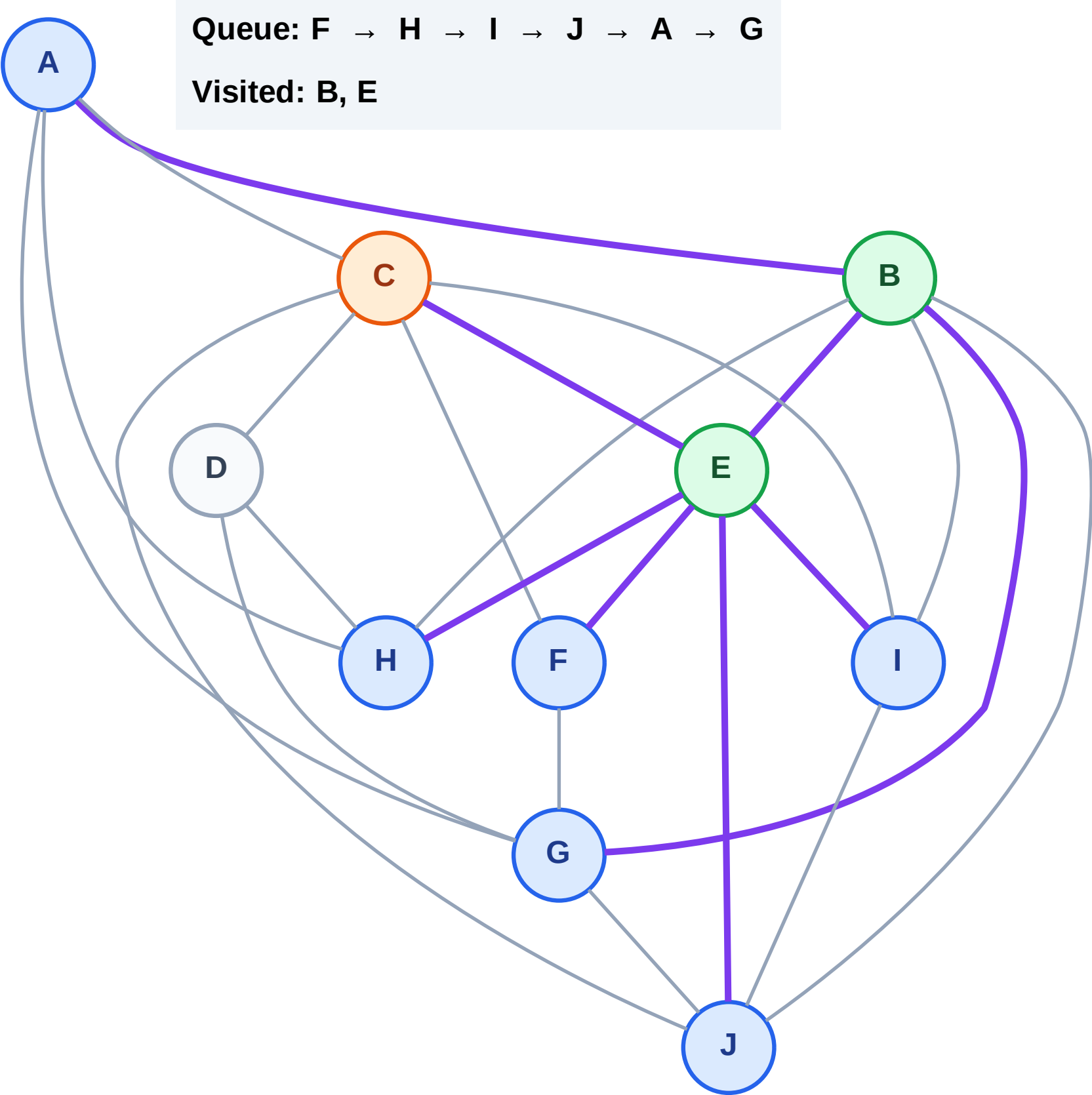
Legend

Complete

Processing

Discovered

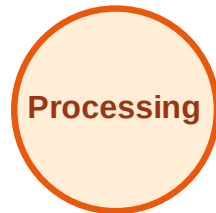
Unseen



# BFS Traversal

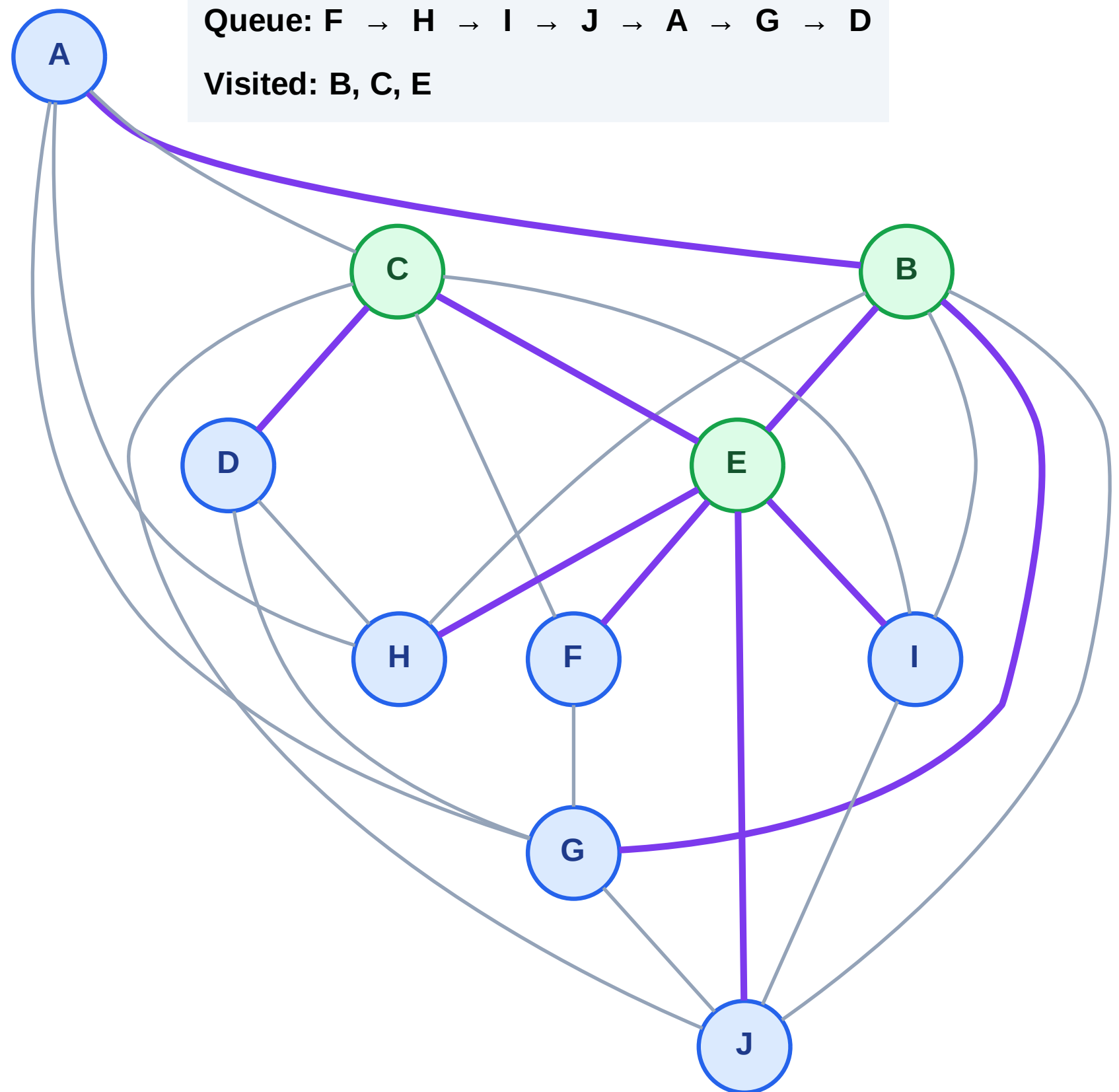
Step 7: Discover D from C

## Legend



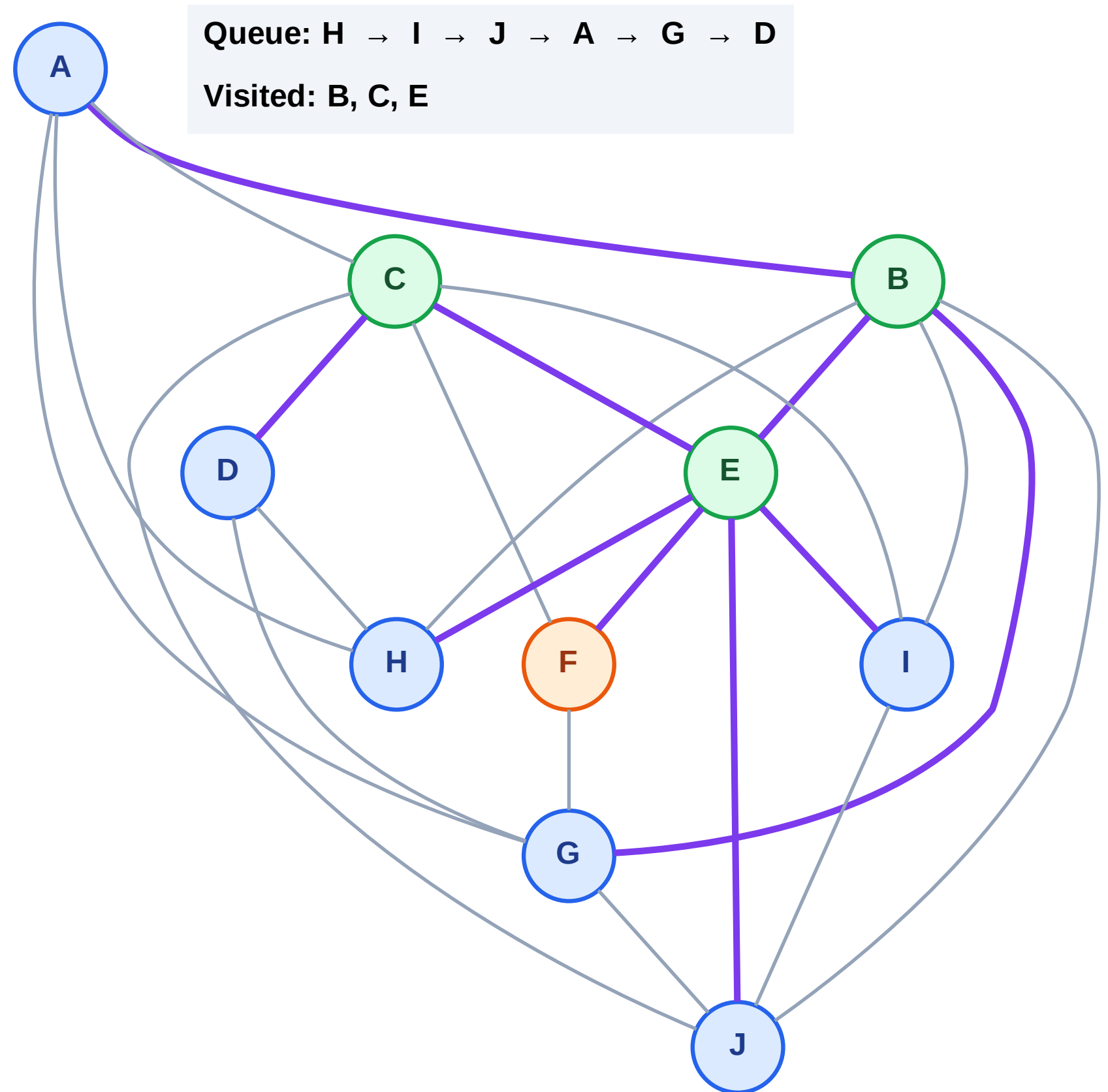
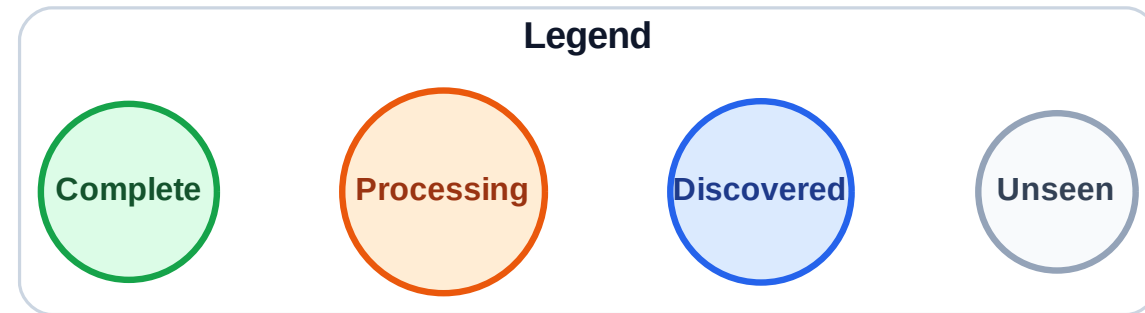
Queue: F → H → I → J → A → G → D

Visited: B, C, E



# BFS Traversal

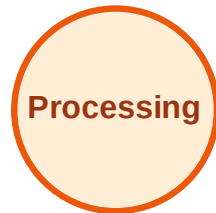
Step 8: Process node F



# BFS Traversal

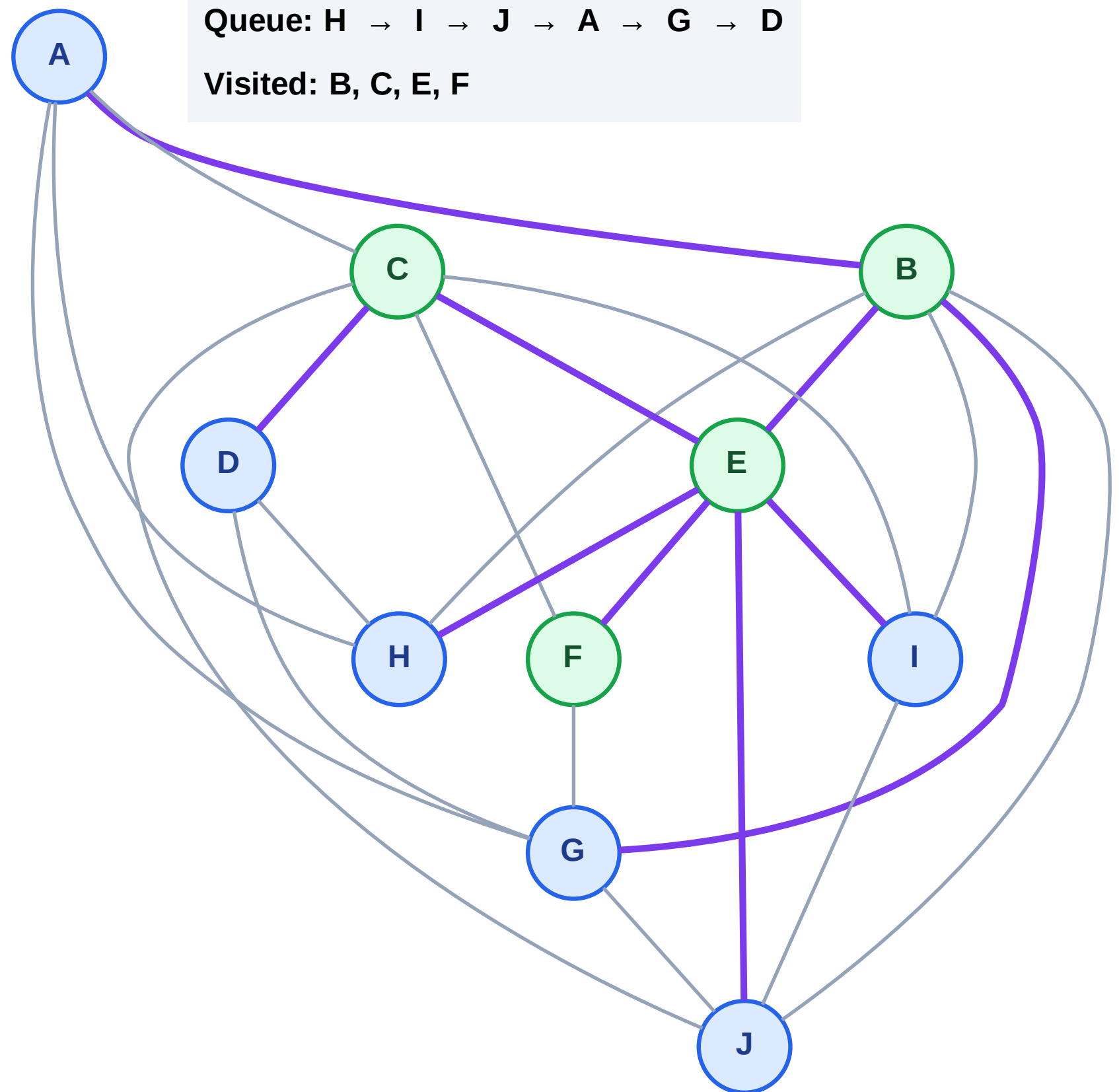
Step 9: No new neighbors from F

## Legend



Queue: H → I → J → A → G → D

Visited: B, C, E, F



# BFS Traversal

Step 10: Process node H

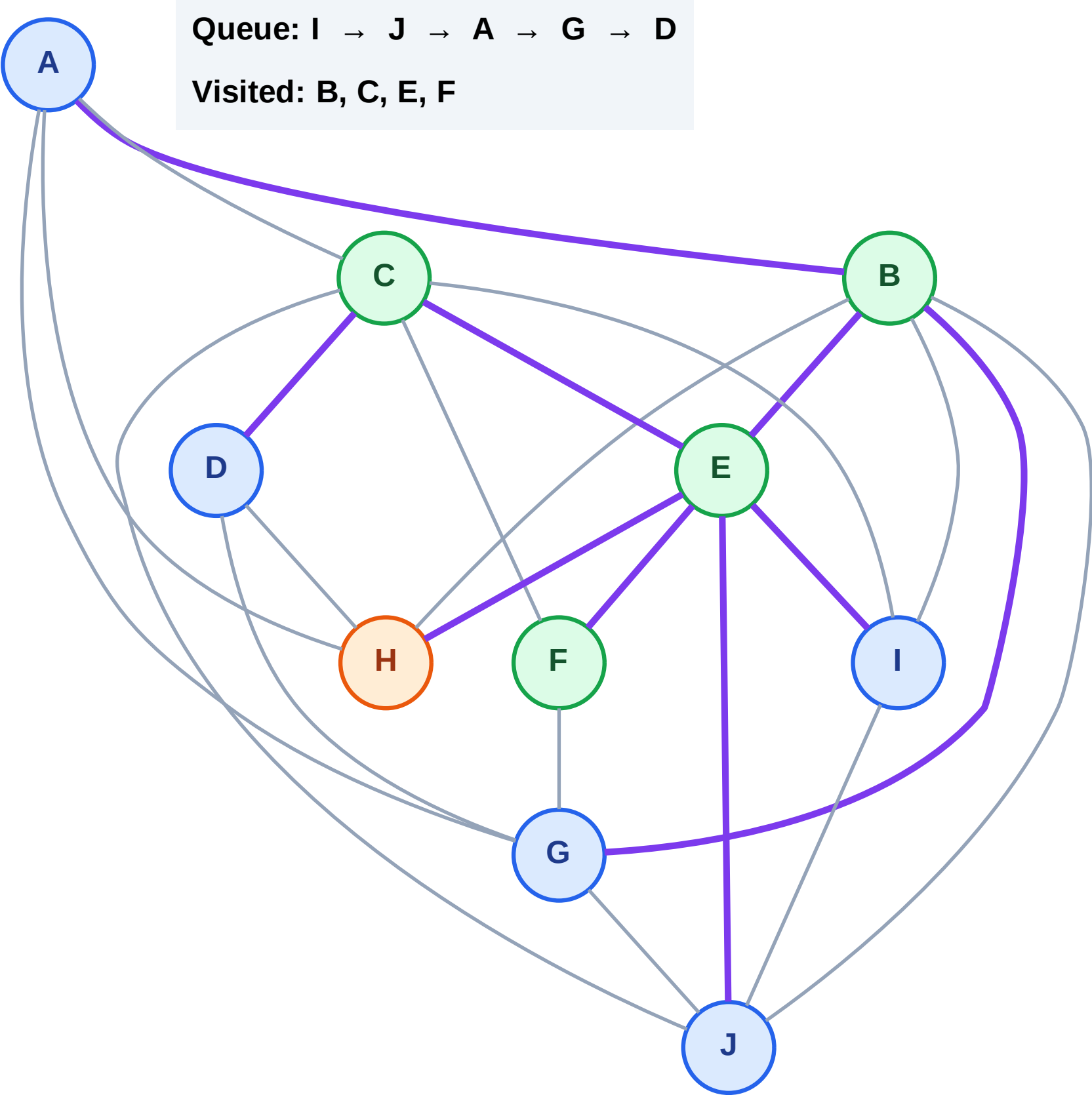
Legend

Complete

Processing

Discovered

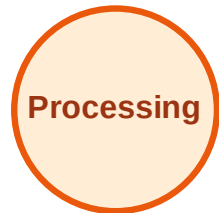
Unseen



# BFS Traversal

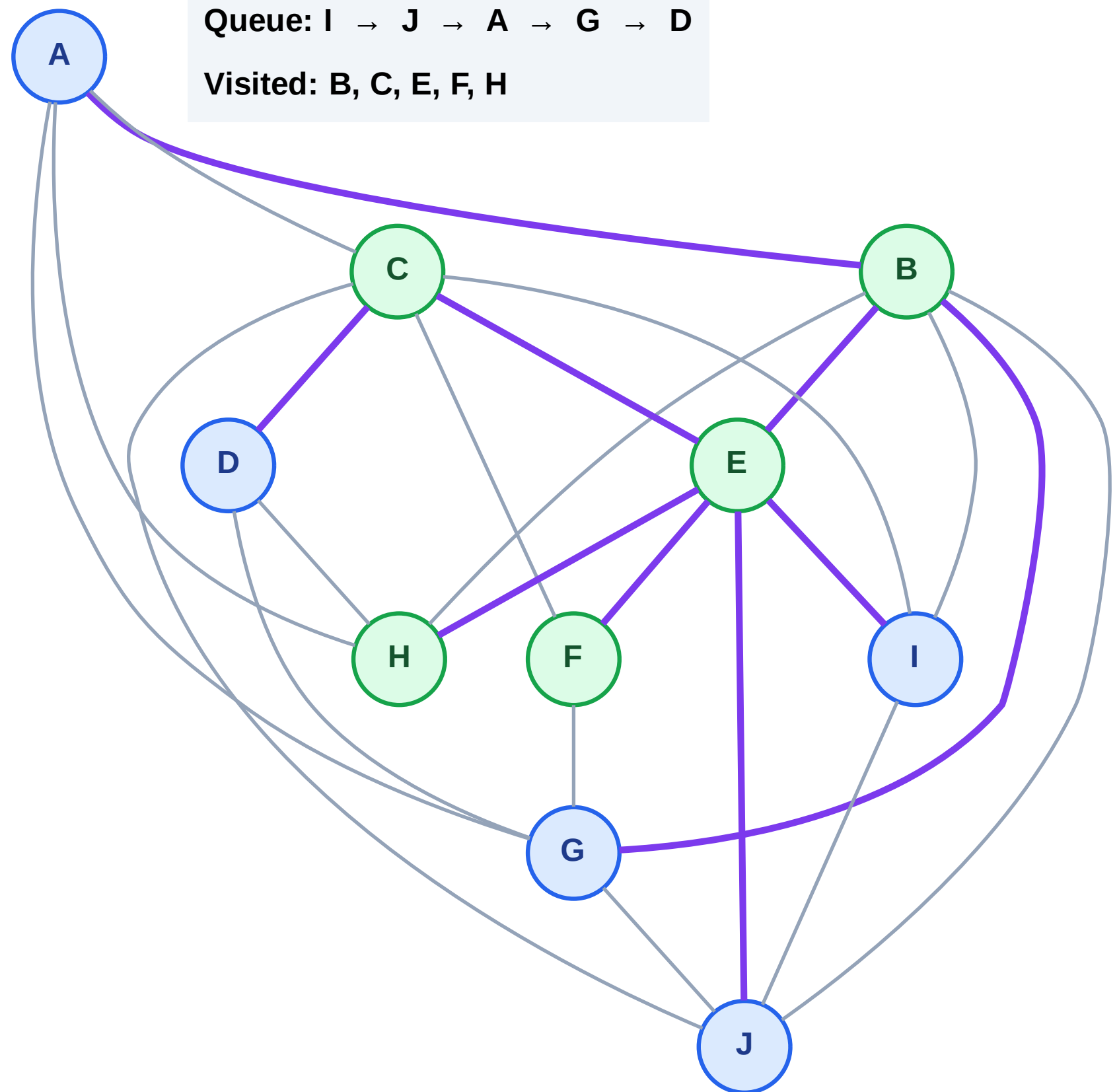
Step 11: No new neighbors from H

## Legend



Queue: I → J → A → G → D

Visited: B, C, E, F, H



# BFS Traversal

Step 12: Process node I

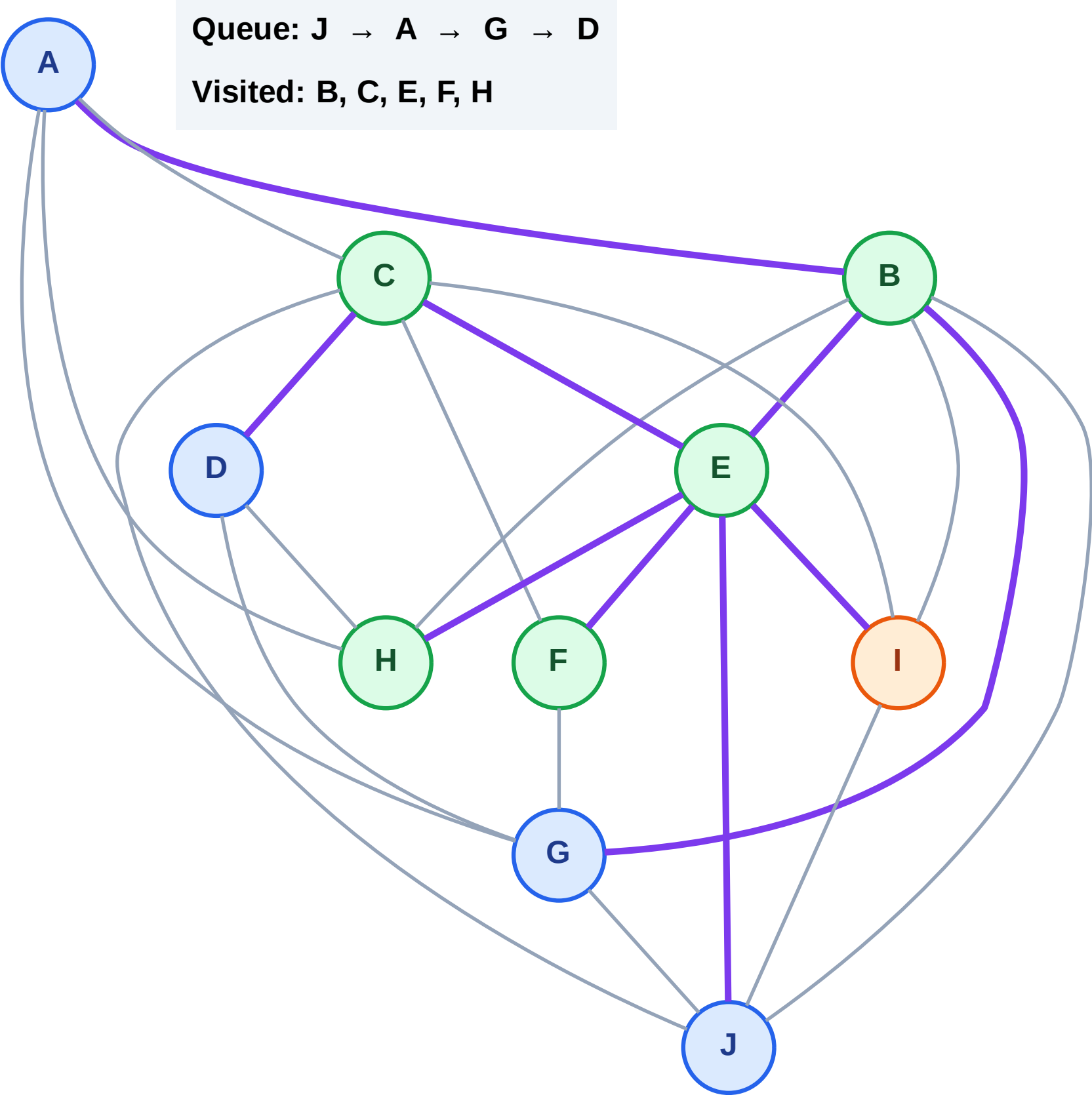
Legend

Complete

Processing

Discovered

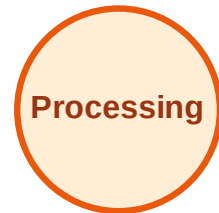
Unseen



# BFS Traversal

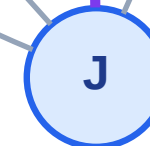
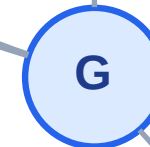
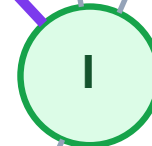
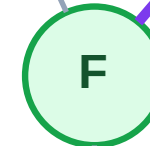
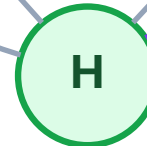
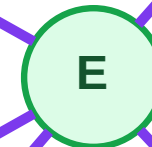
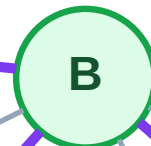
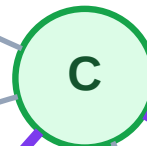
Step 13: No new neighbors from I

## Legend



Queue: J → A → G → D

Visited: B, C, E, F, H, I



# BFS Traversal

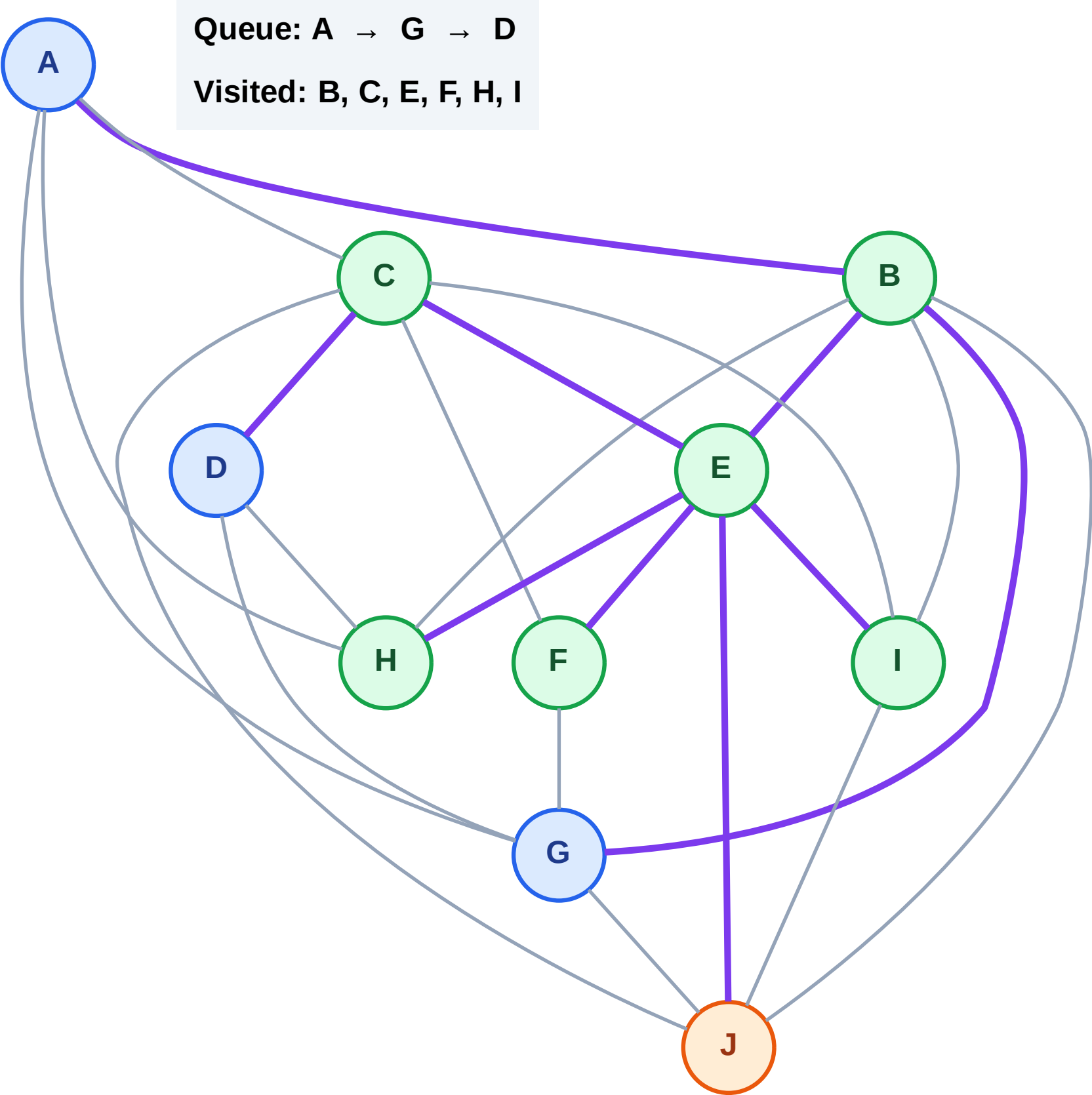
Step 14: Process node J

Complete

Processing

Discovered

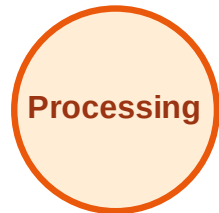
Unseen



# BFS Traversal

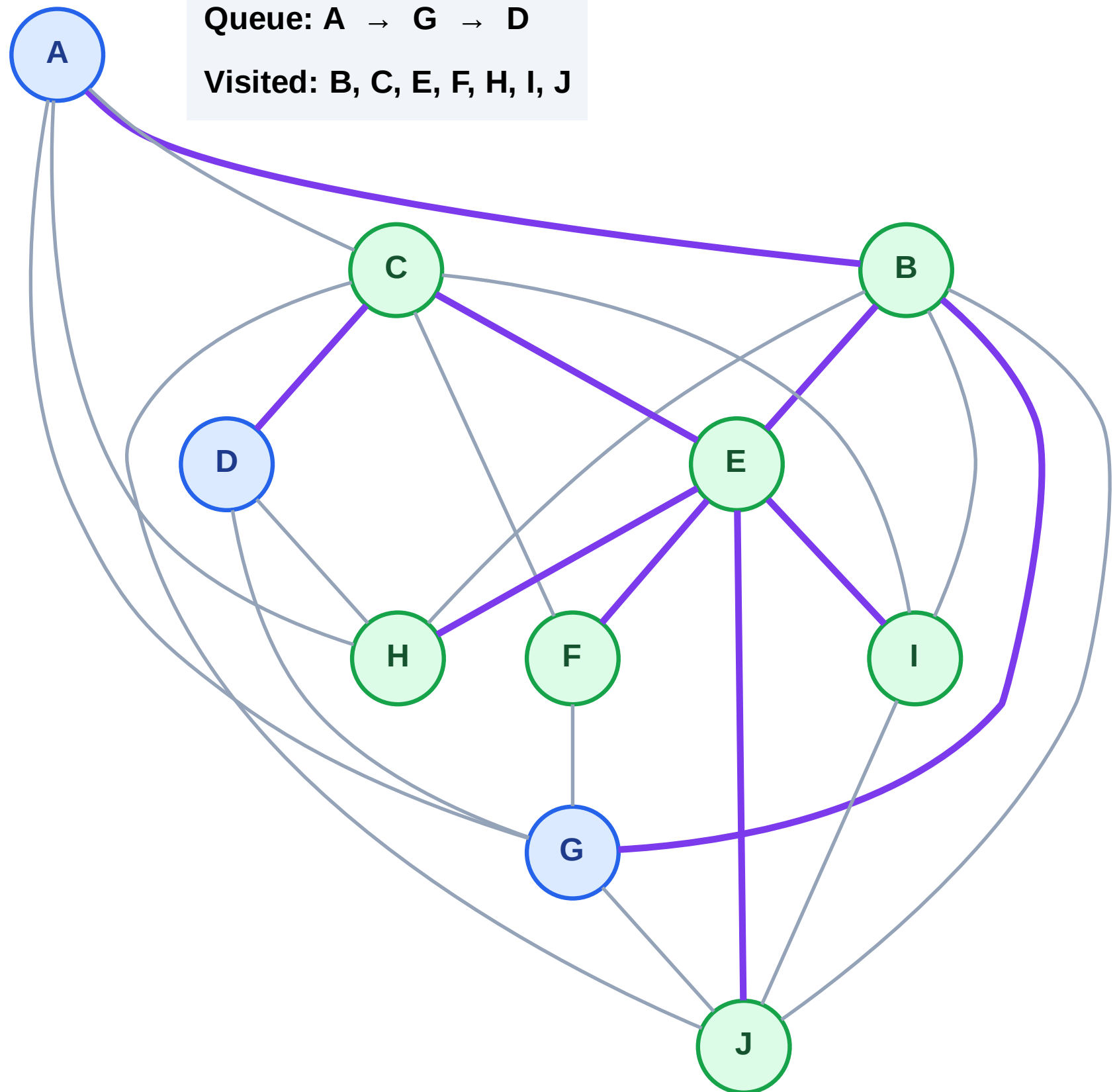
Step 15: No new neighbors from J

## Legend



Queue: A → G → D

Visited: B, C, E, F, H, I, J



# BFS Traversal

Step 16: Process node A

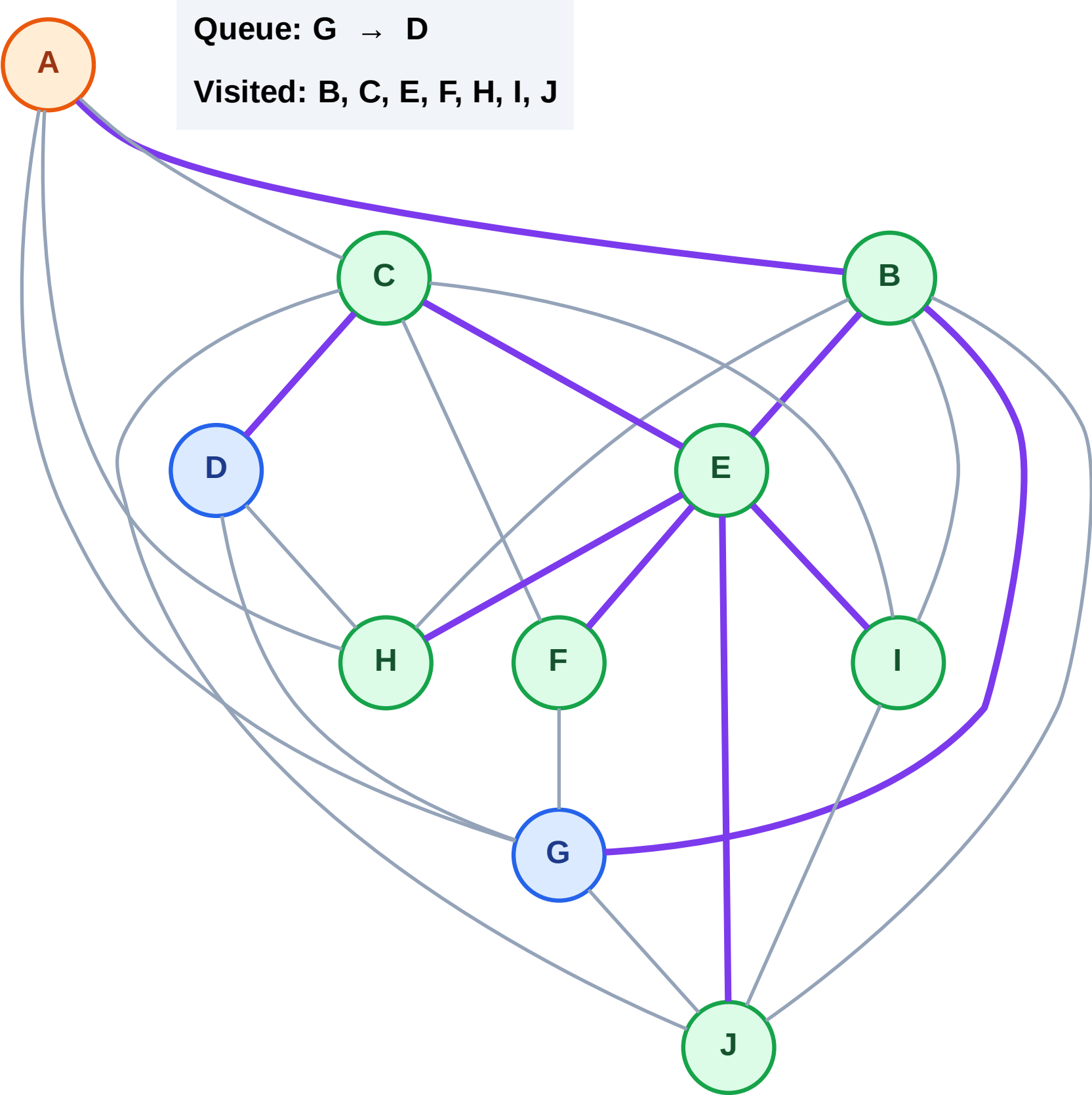
Legend

Complete

Processing

Discovered

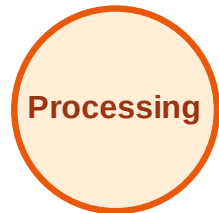
Unseen



# BFS Traversal

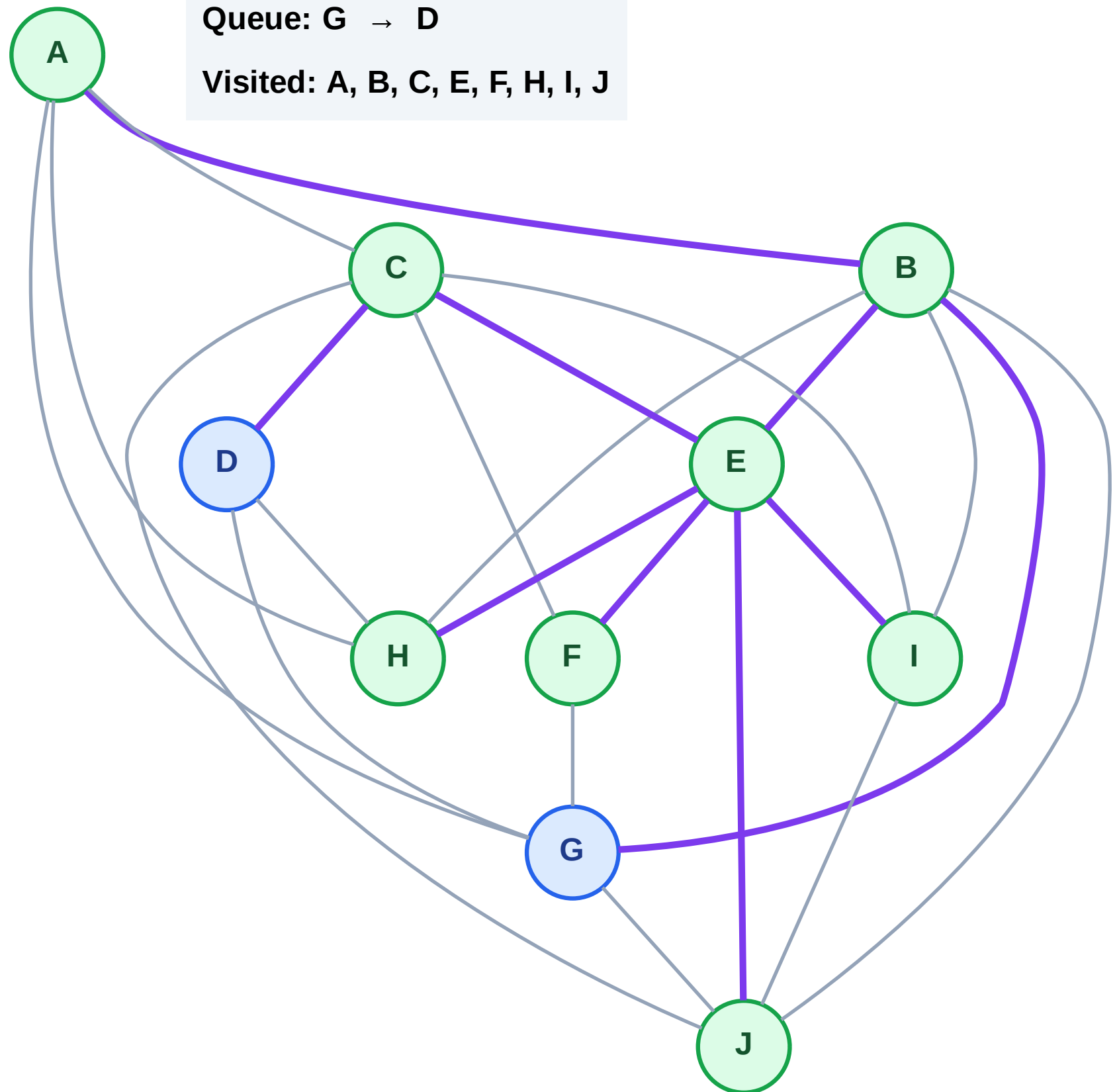
Step 17: No new neighbors from A

## Legend



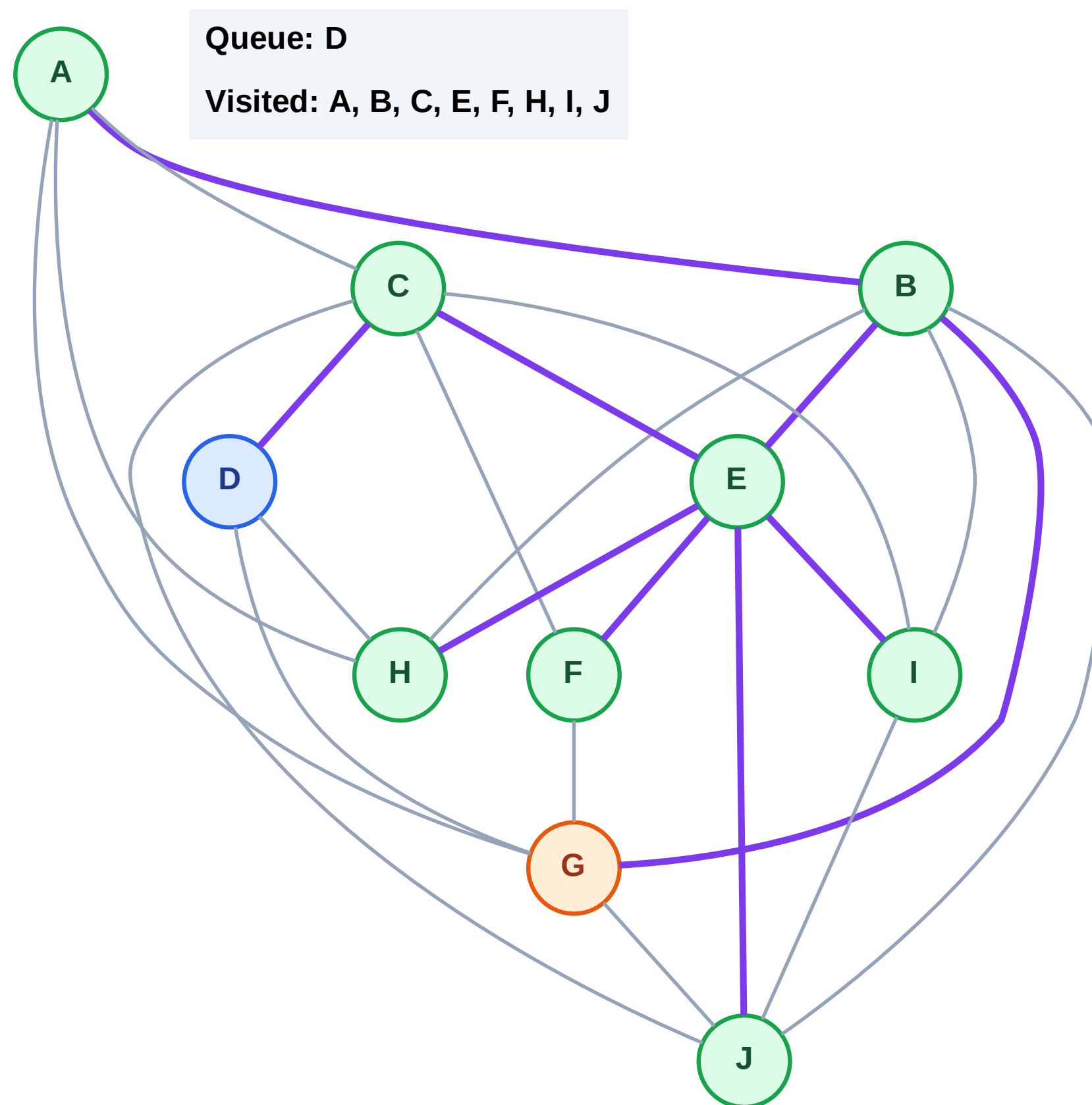
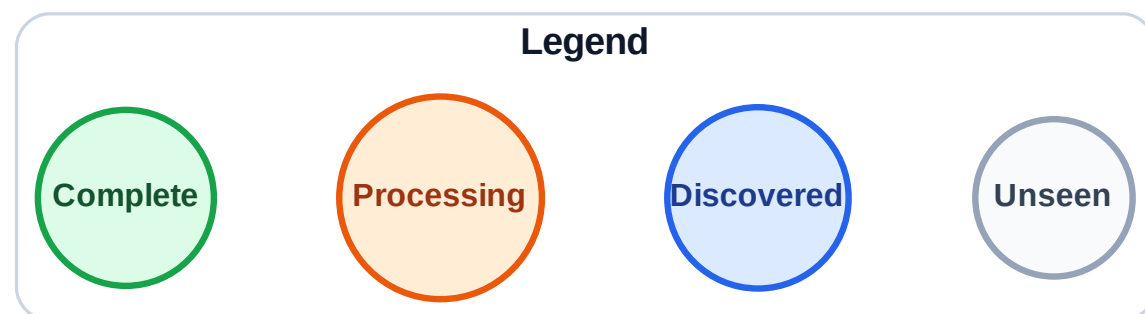
Queue: G → D

Visited: A, B, C, E, F, H, I, J



# BFS Traversal

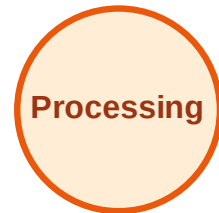
Step 18: Process node G



# BFS Traversal

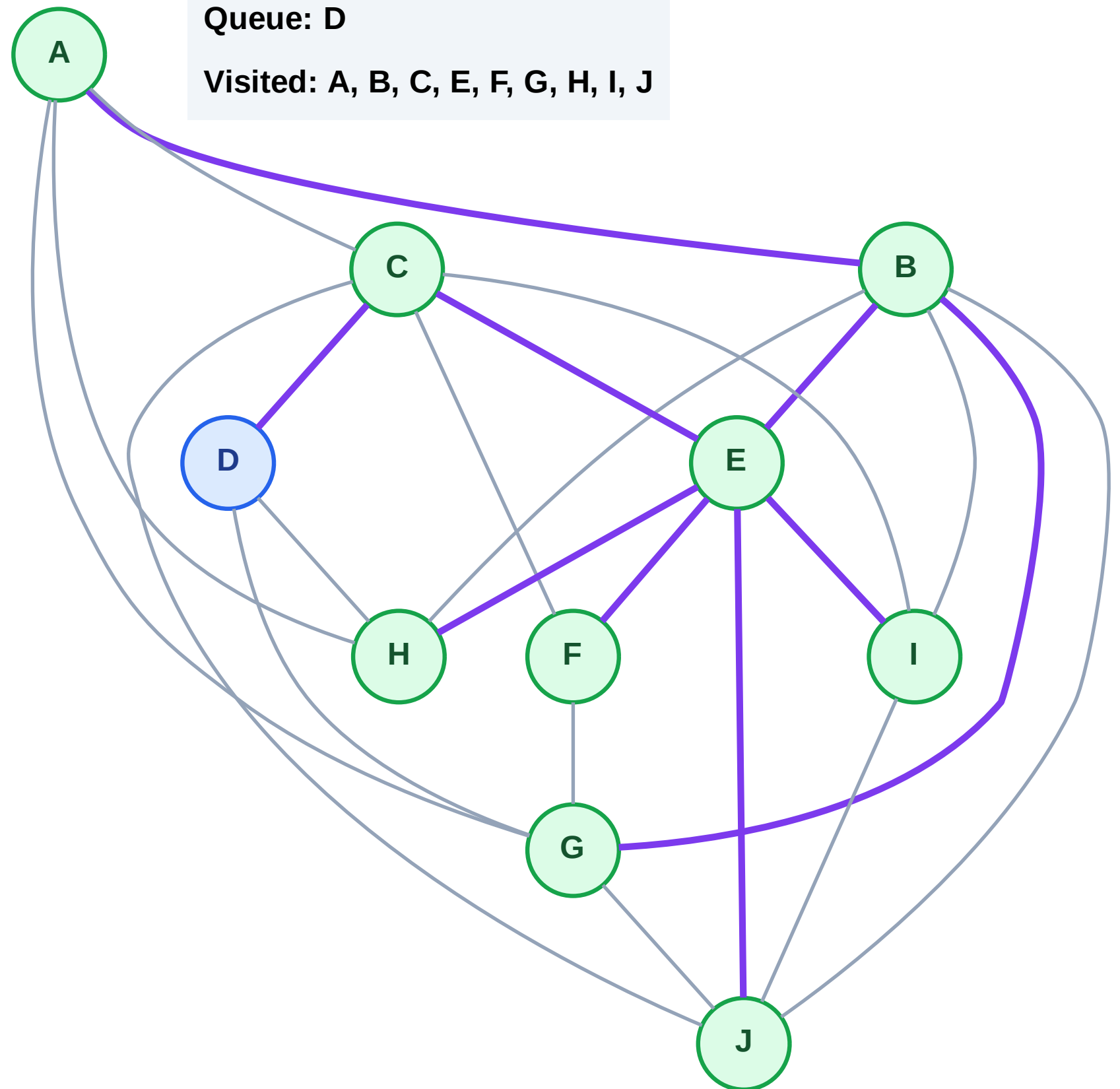
Step 19: No new neighbors from G

## Legend



Queue: D

Visited: A, B, C, E, F, G, H, I, J



# BFS Traversal

Step 20: Process node D

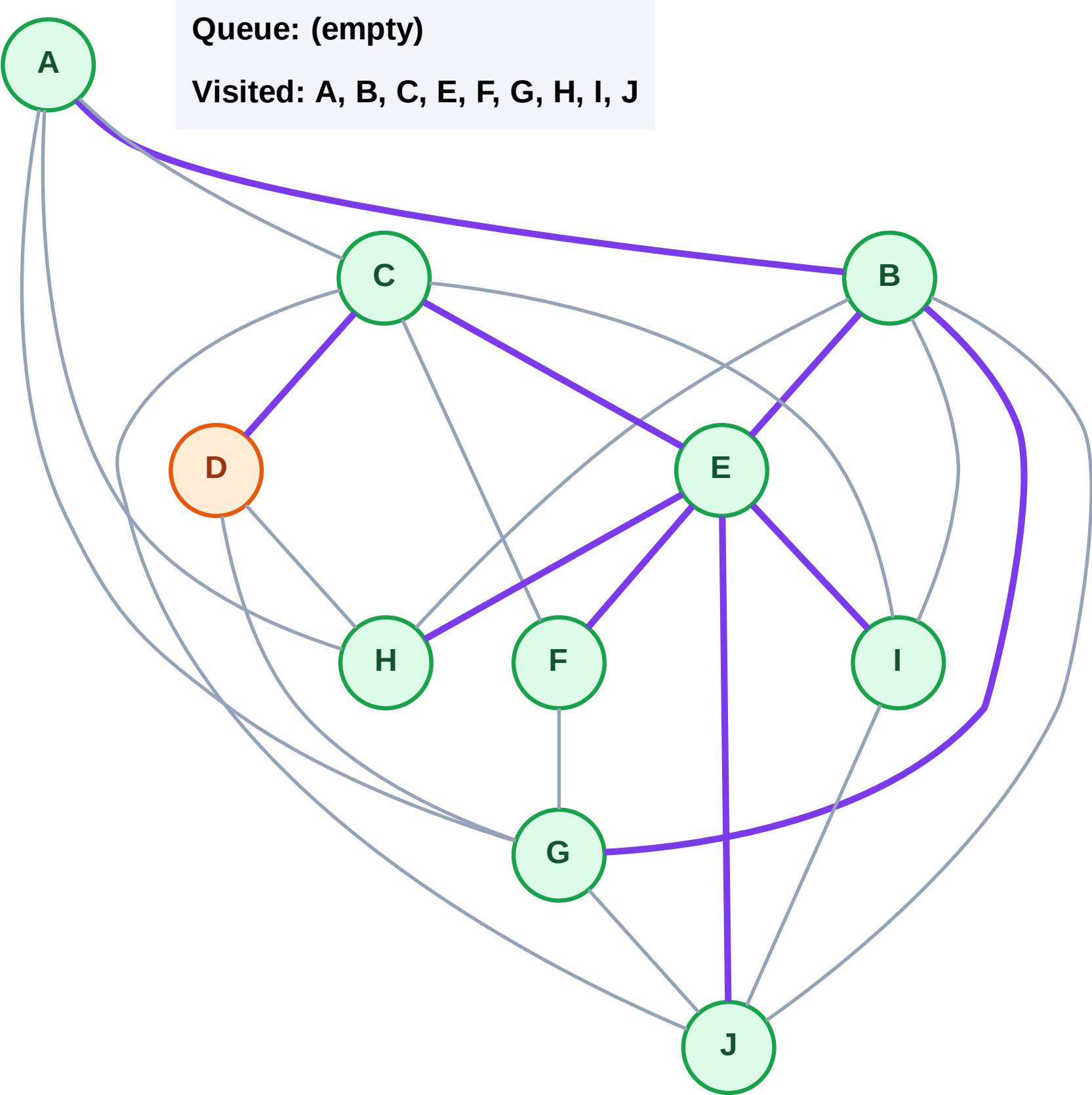
Legend

Complete

Processing

Discovered

Unseen



# BFS Traversal

Step 21: No new neighbors from D

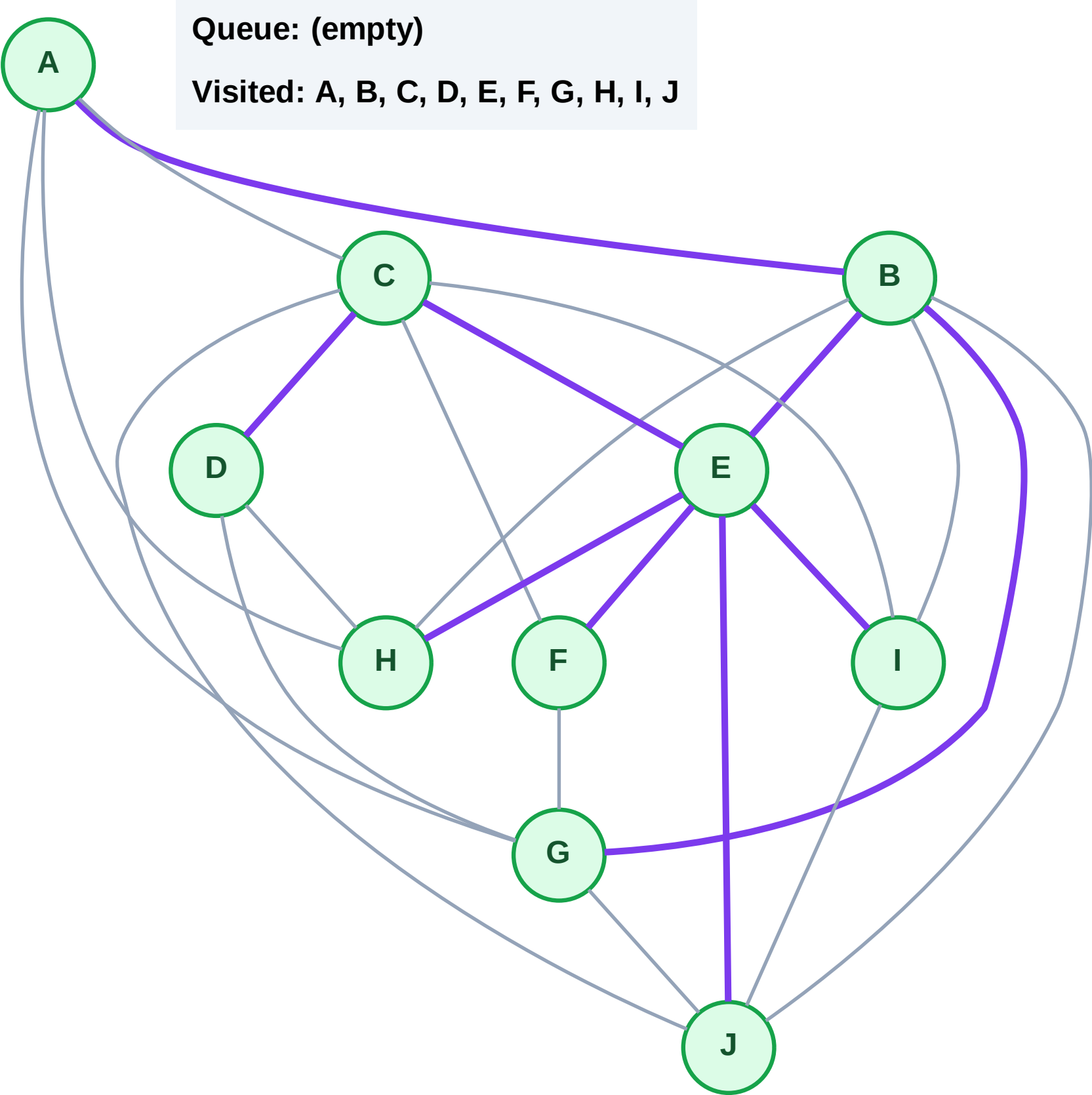
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 1: Start at E

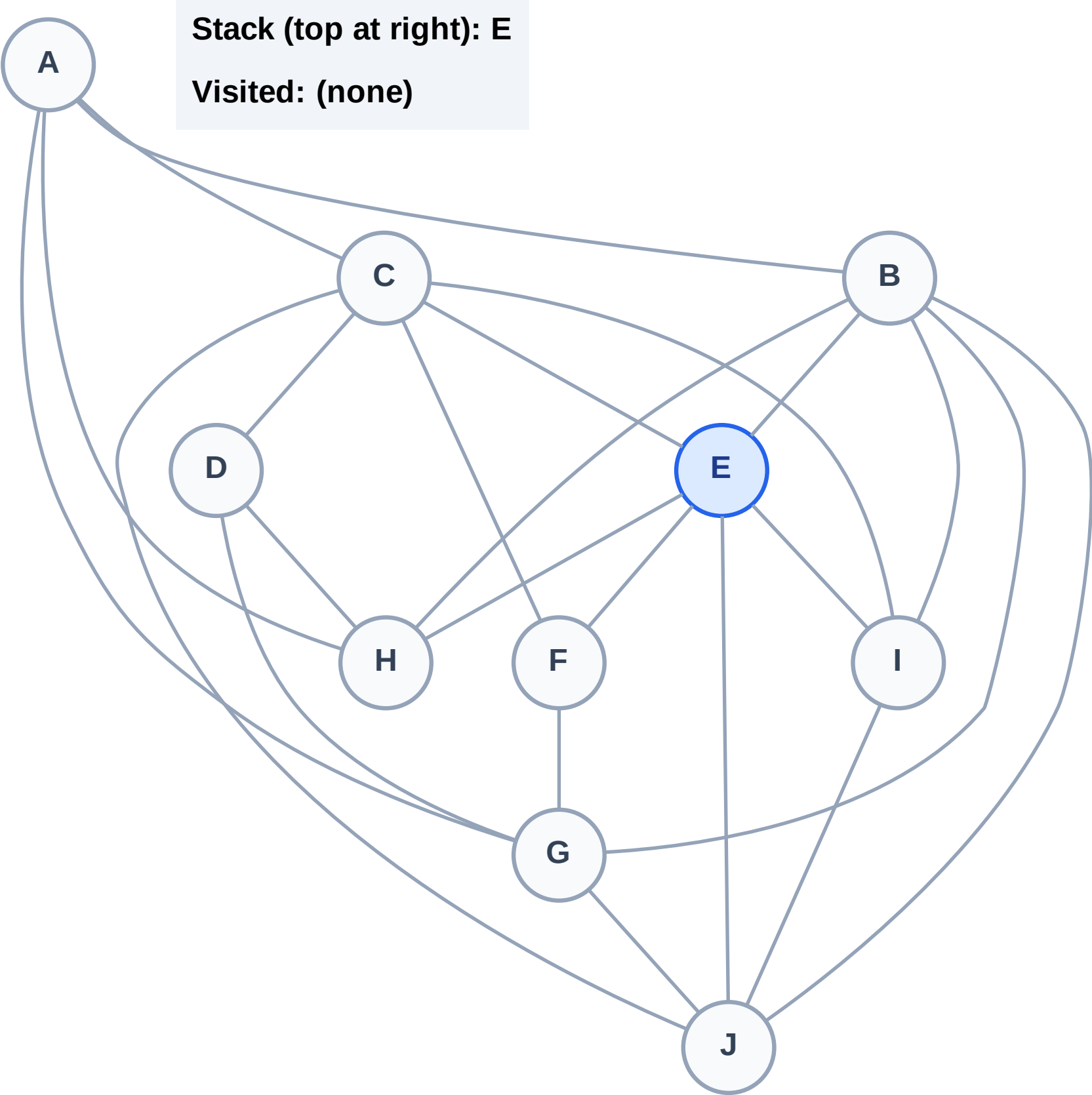
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 2: Process node E

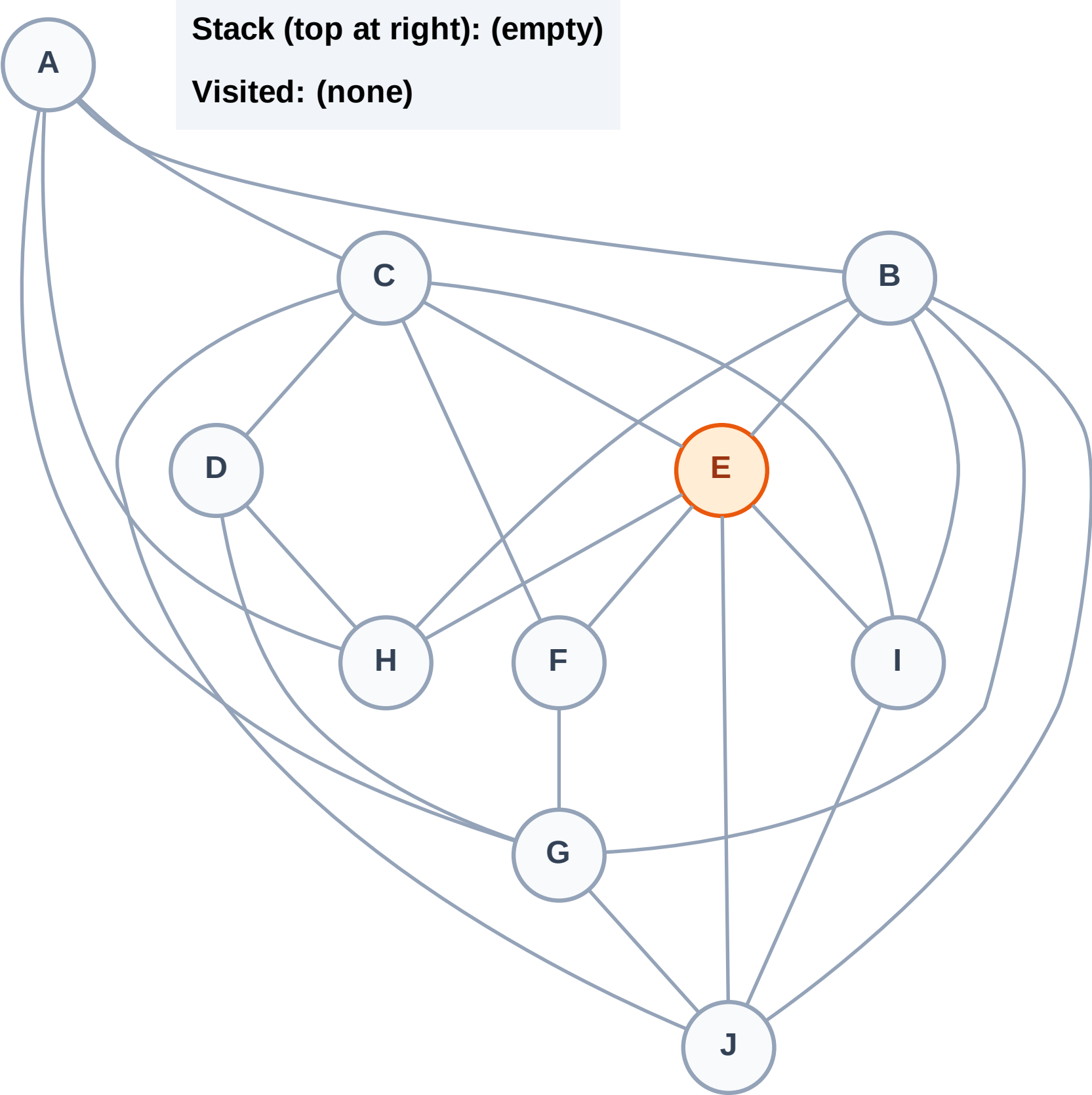
Legend

Complete

Processing

Discovered

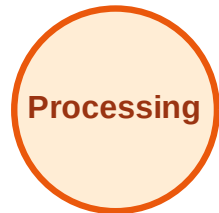
Unseen



# DFS Traversal

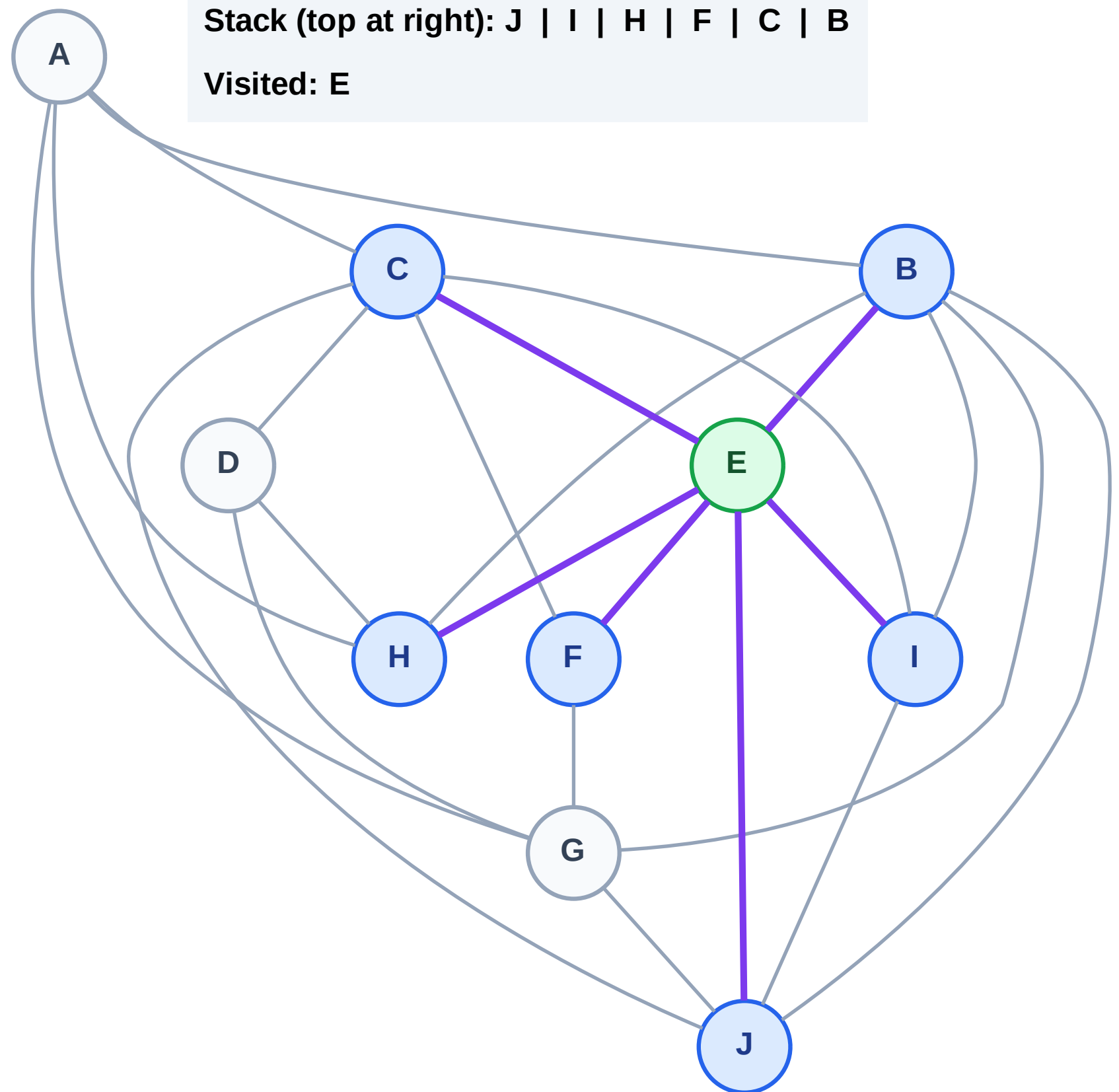
Step 3: Discover J, I, H, F, C, B from E

## Legend



Stack (top at right): J | I | H | F | C | B

Visited: E



# DFS Traversal

Step 4: Process node B

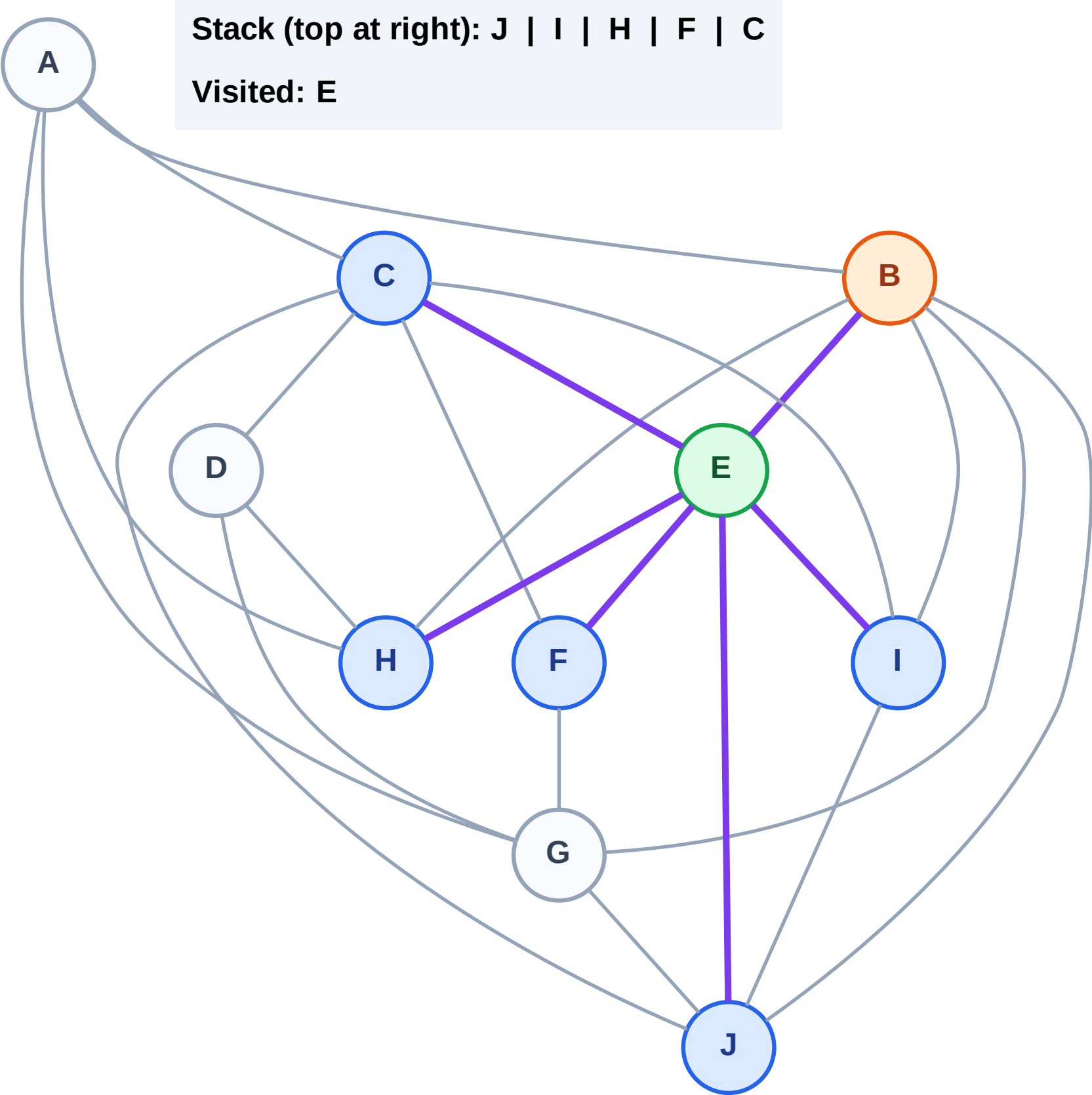
Legend

Complete

Processing

Discovered

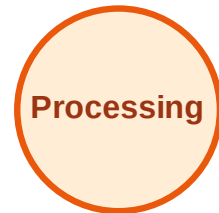
Unseen



# DFS Traversal

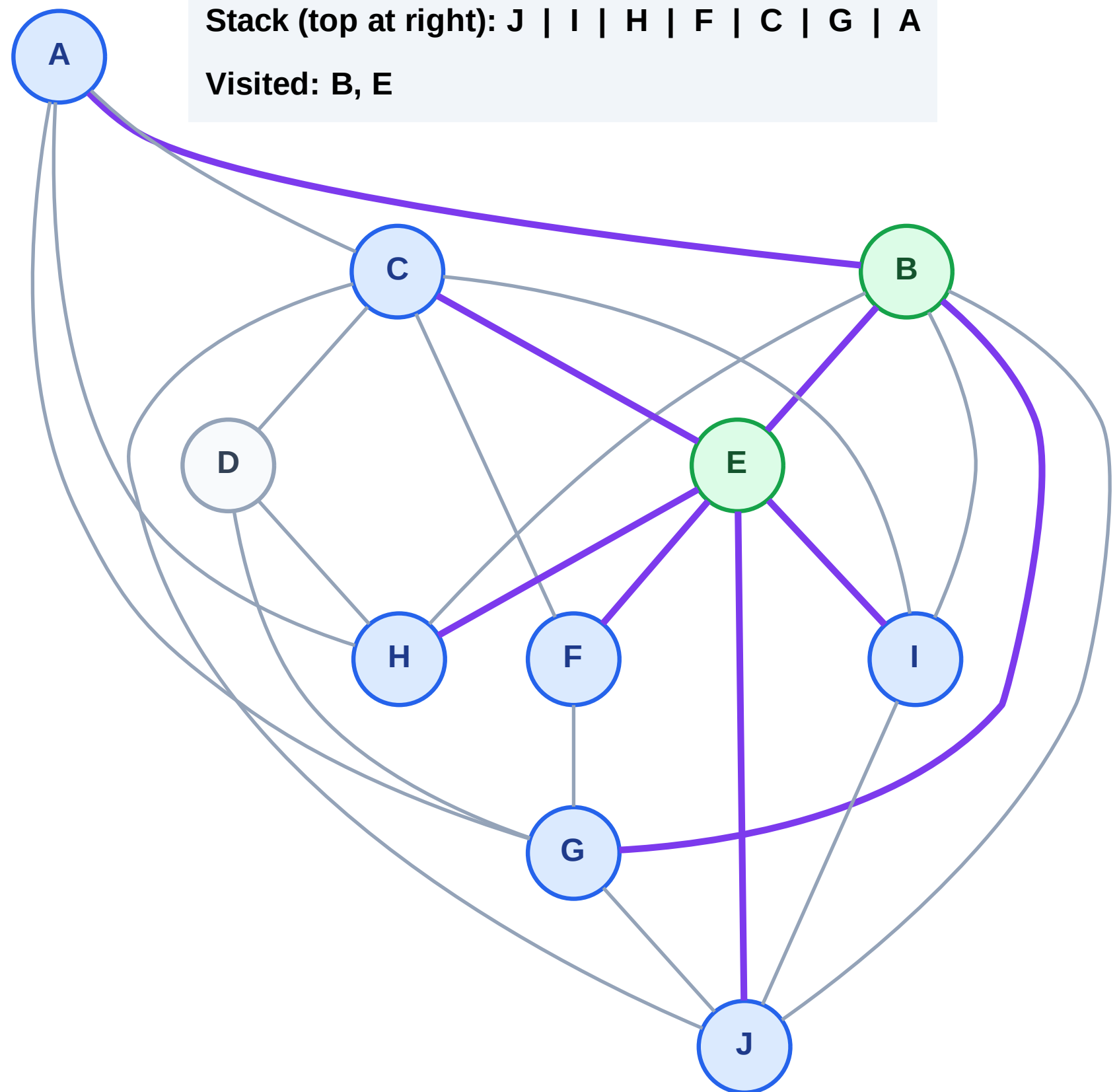
Step 5: Discover G, A from B

## Legend



Stack (top at right): J | I | H | F | C | G | A

Visited: B, E



# DFS Traversal

Step 6: Process node A

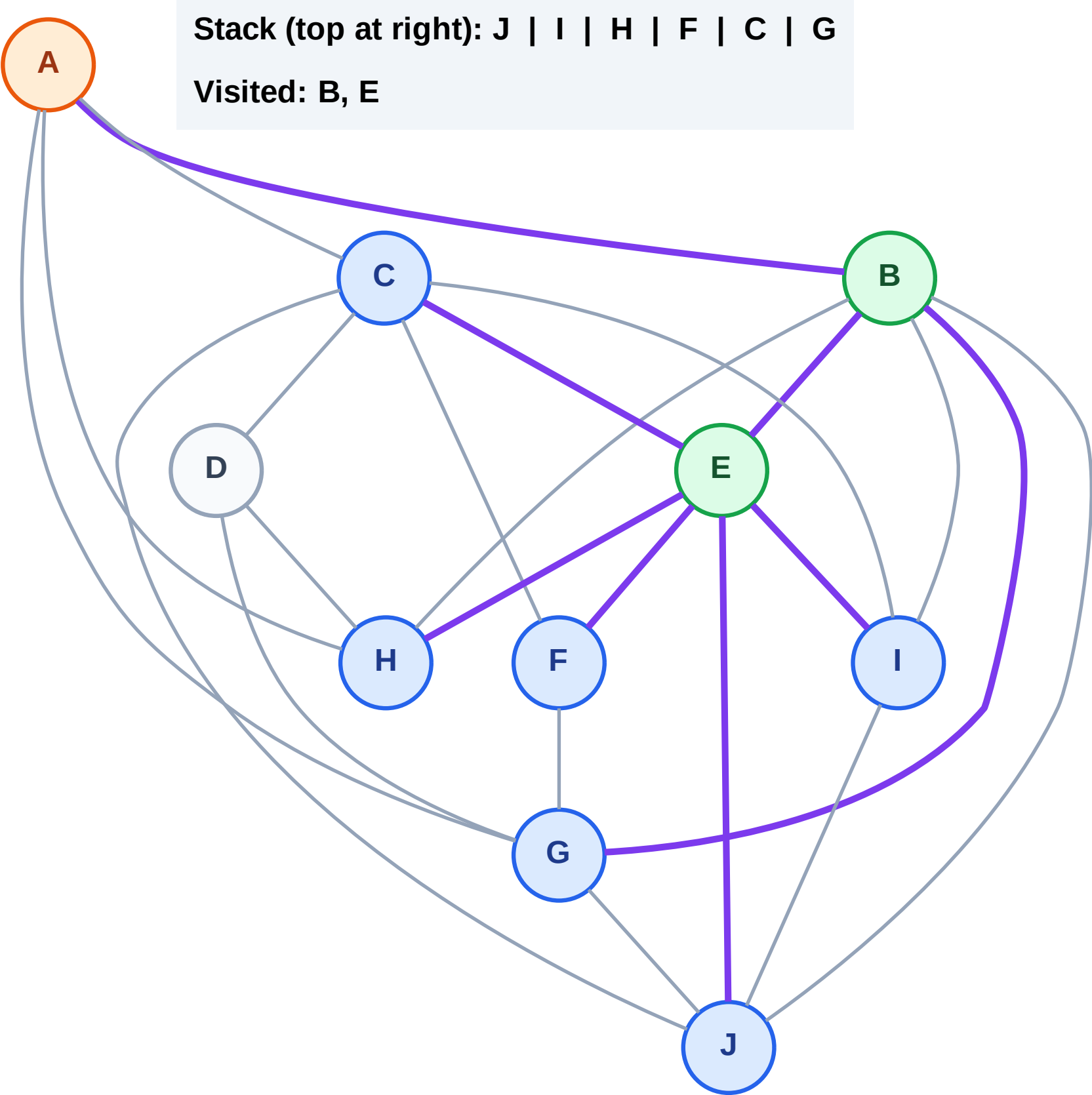
Legend

Complete

Processing

Discovered

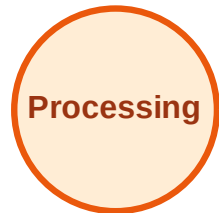
Unseen



# DFS Traversal

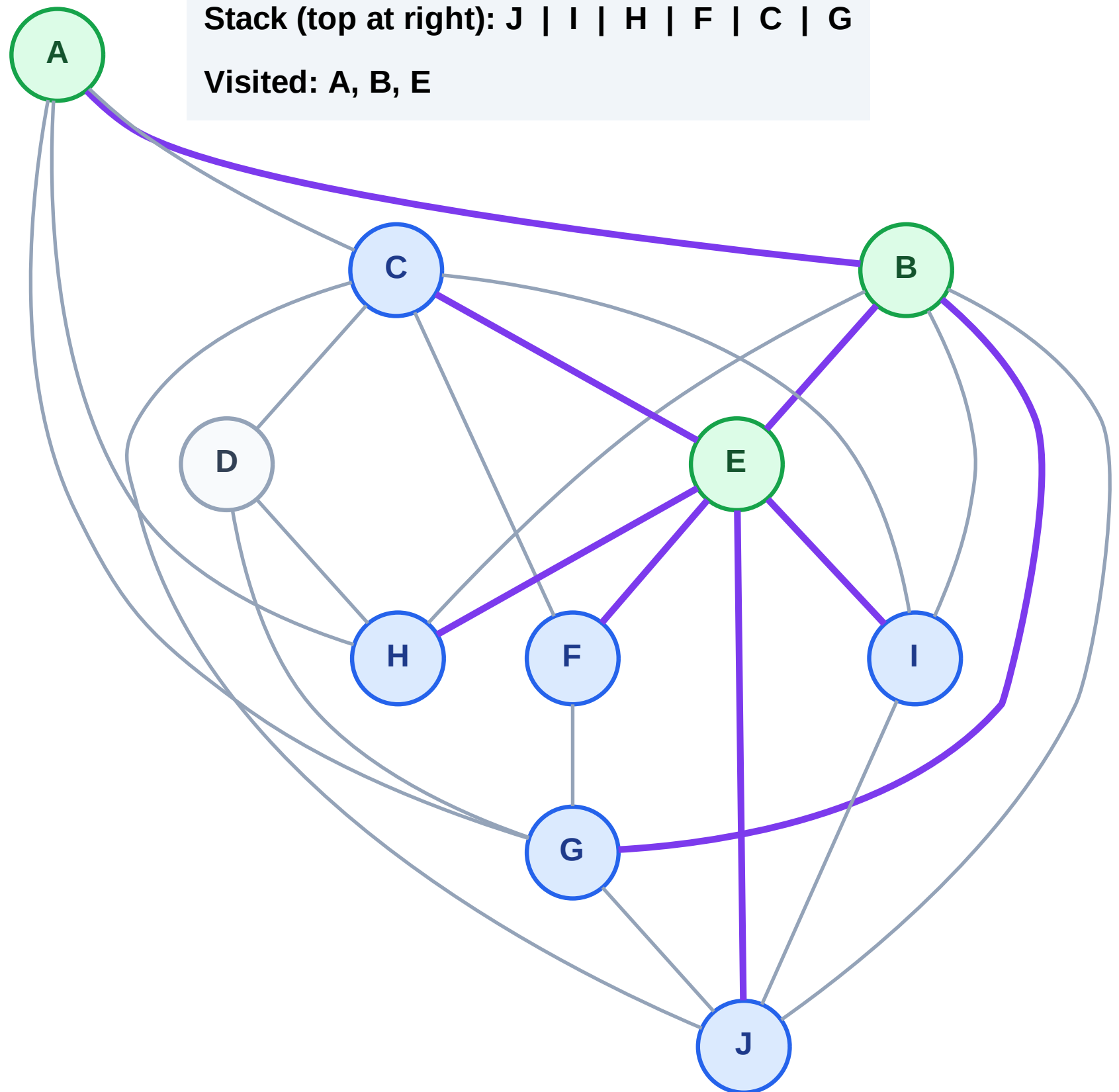
Step 7: No new neighbors from A

## Legend

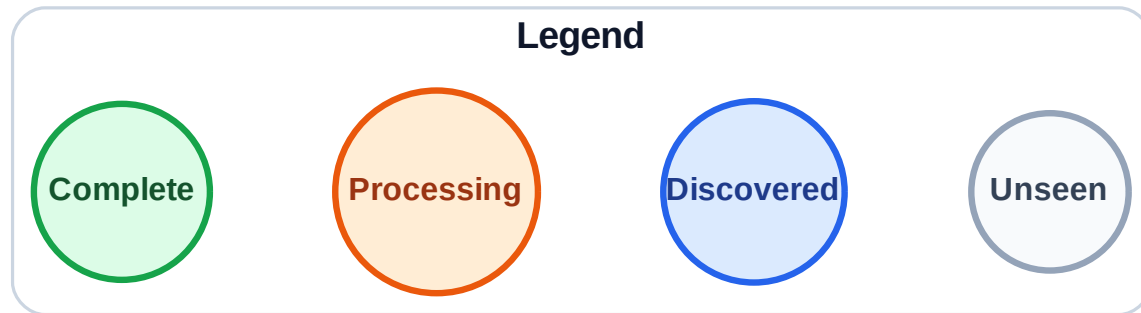


Stack (top at right): J | I | H | F | C | G

Visited: A, B, E

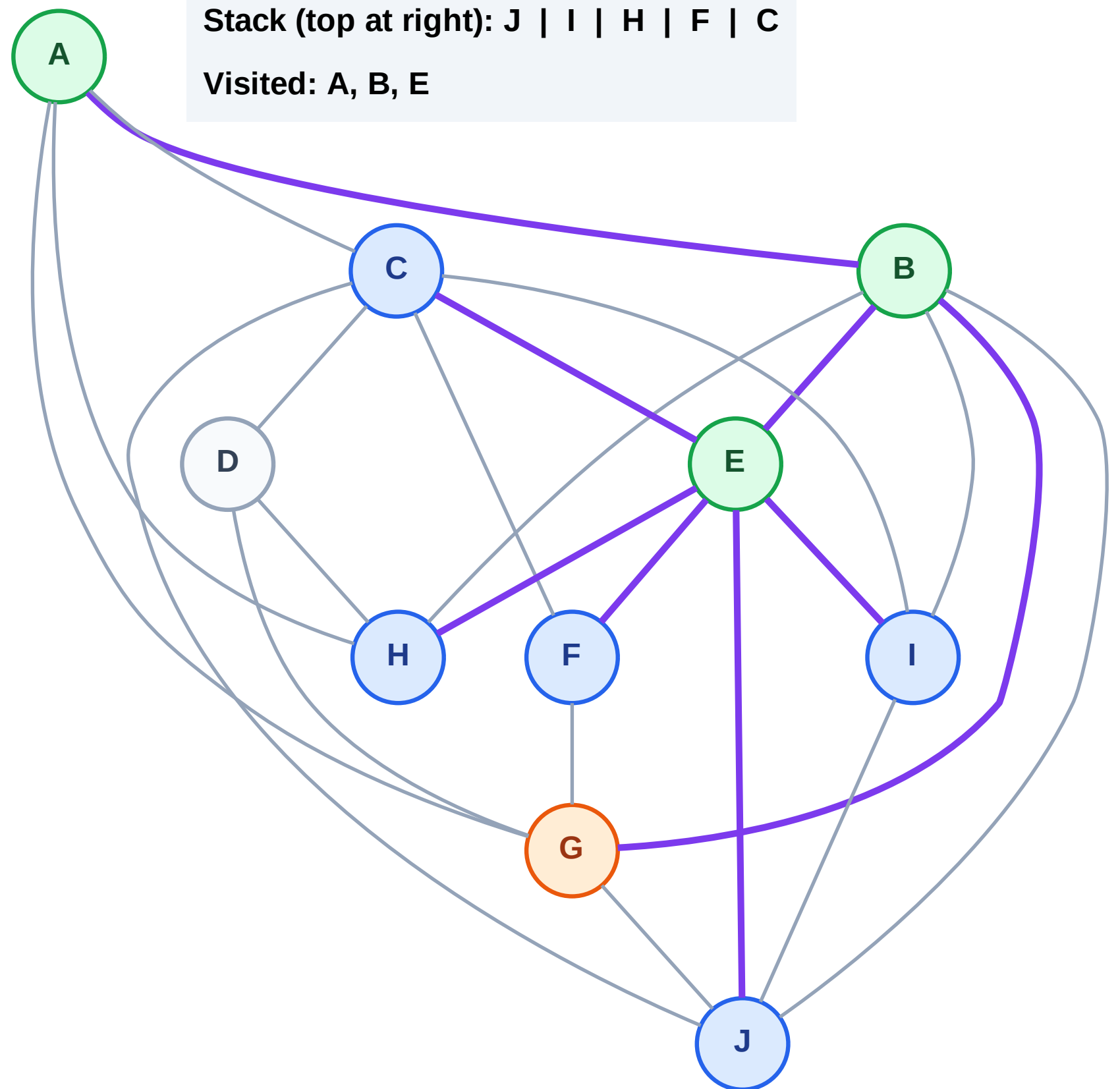


### Step 8: Process node G



**Stack (top at right): J | I | H | F | C**

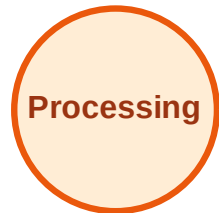
**Visited: A, B, E**



# DFS Traversal

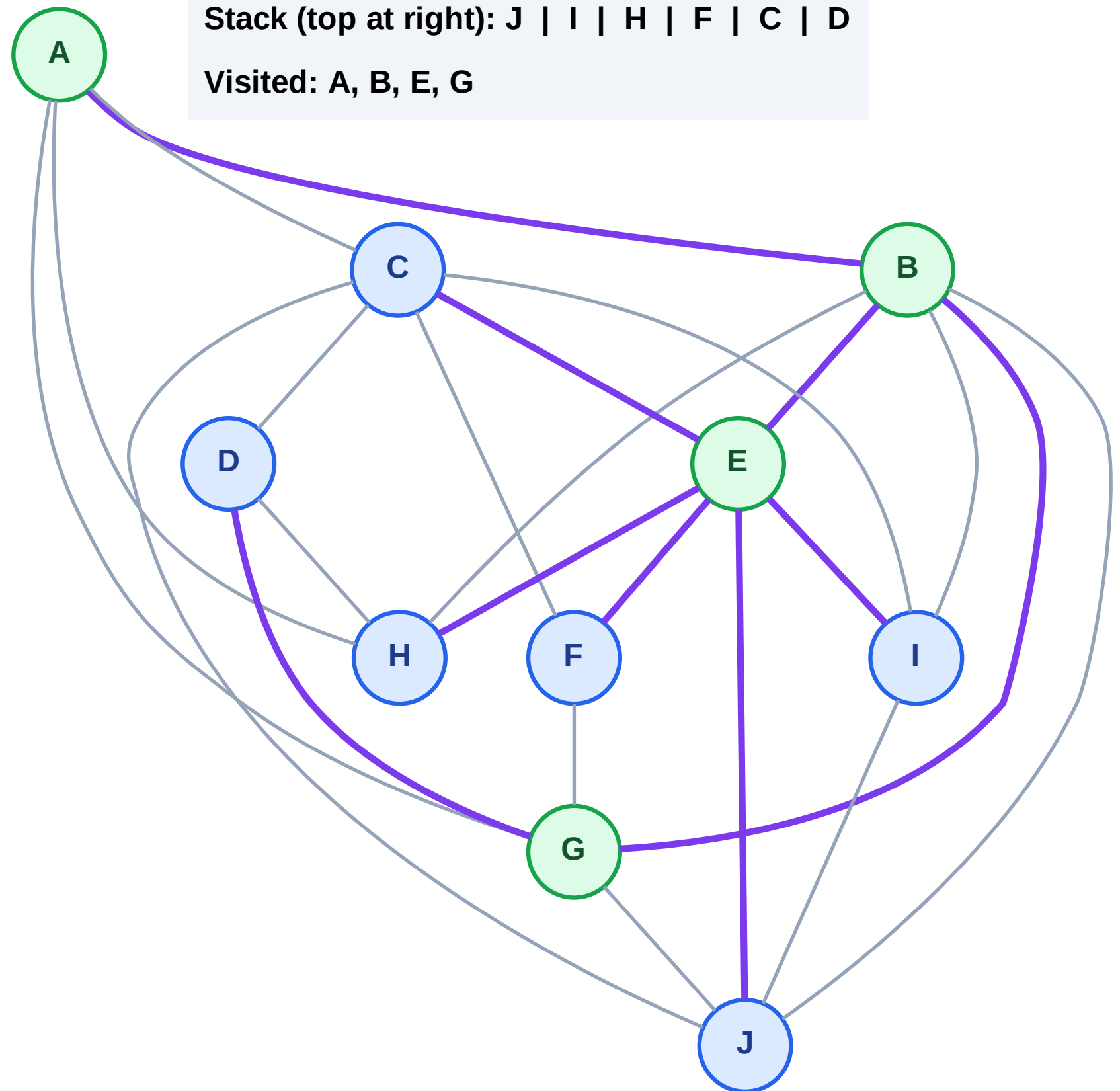
Step 9: Discover D from G

## Legend



Stack (top at right): J | I | H | F | C | D

Visited: A, B, E, G



# DFS Traversal

Step 10: Process node D

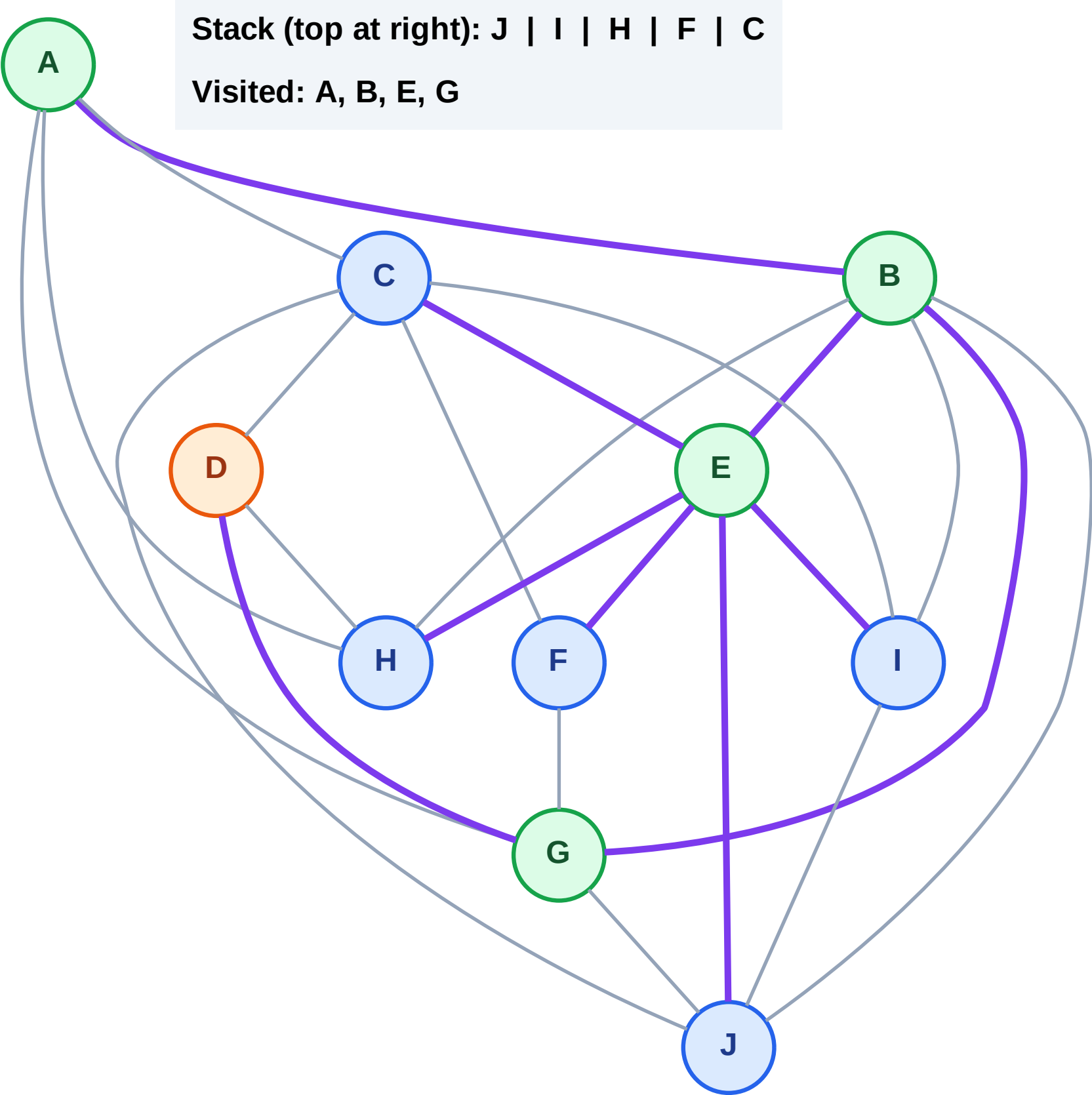
Legend

Complete

Processing

Discovered

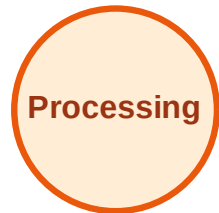
Unseen



# DFS Traversal

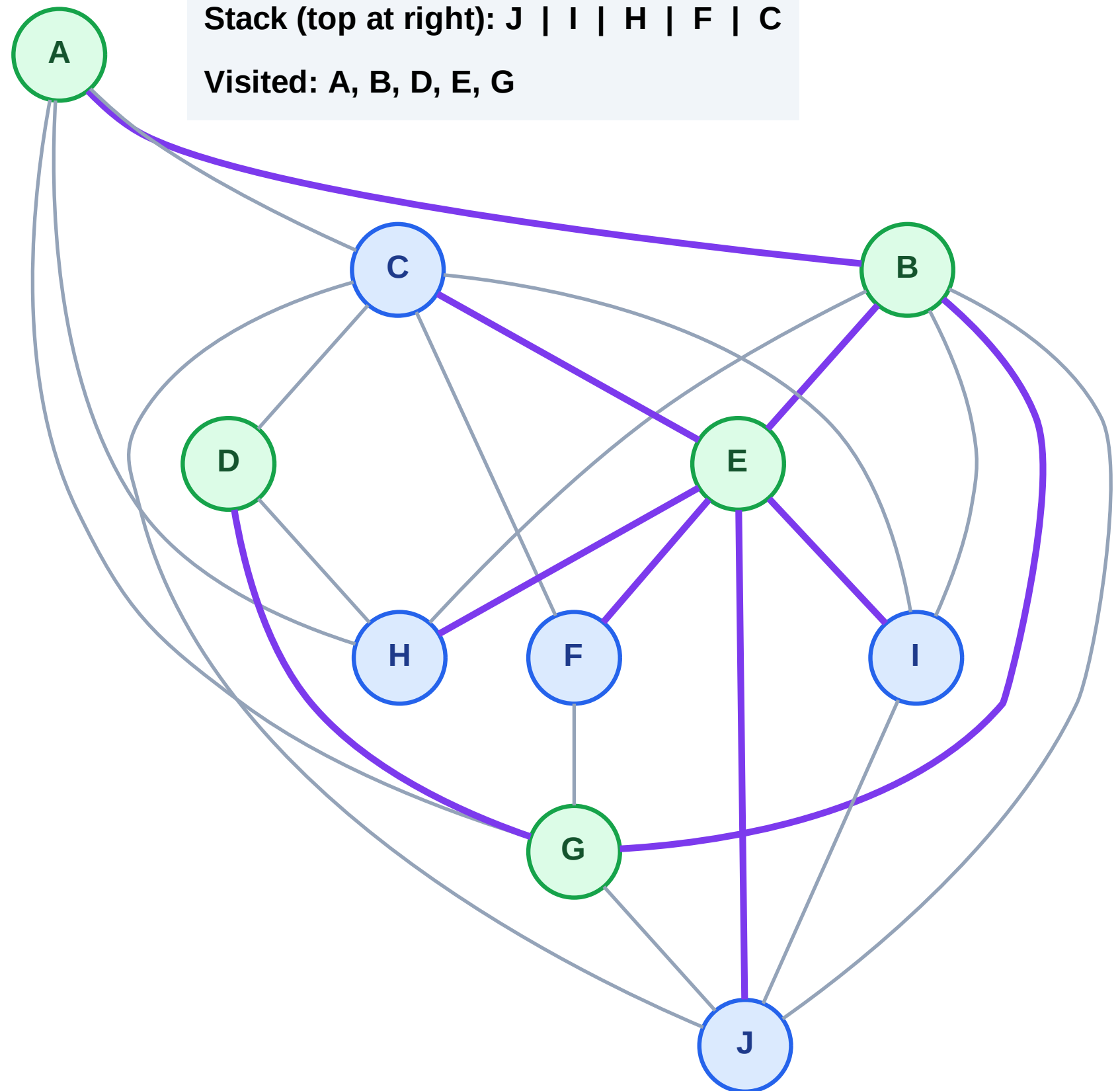
Step 11: No new neighbors from D

## Legend



Stack (top at right): J | I | H | F | C

Visited: A, B, D, E, G



# DFS Traversal

Step 12: Process node C

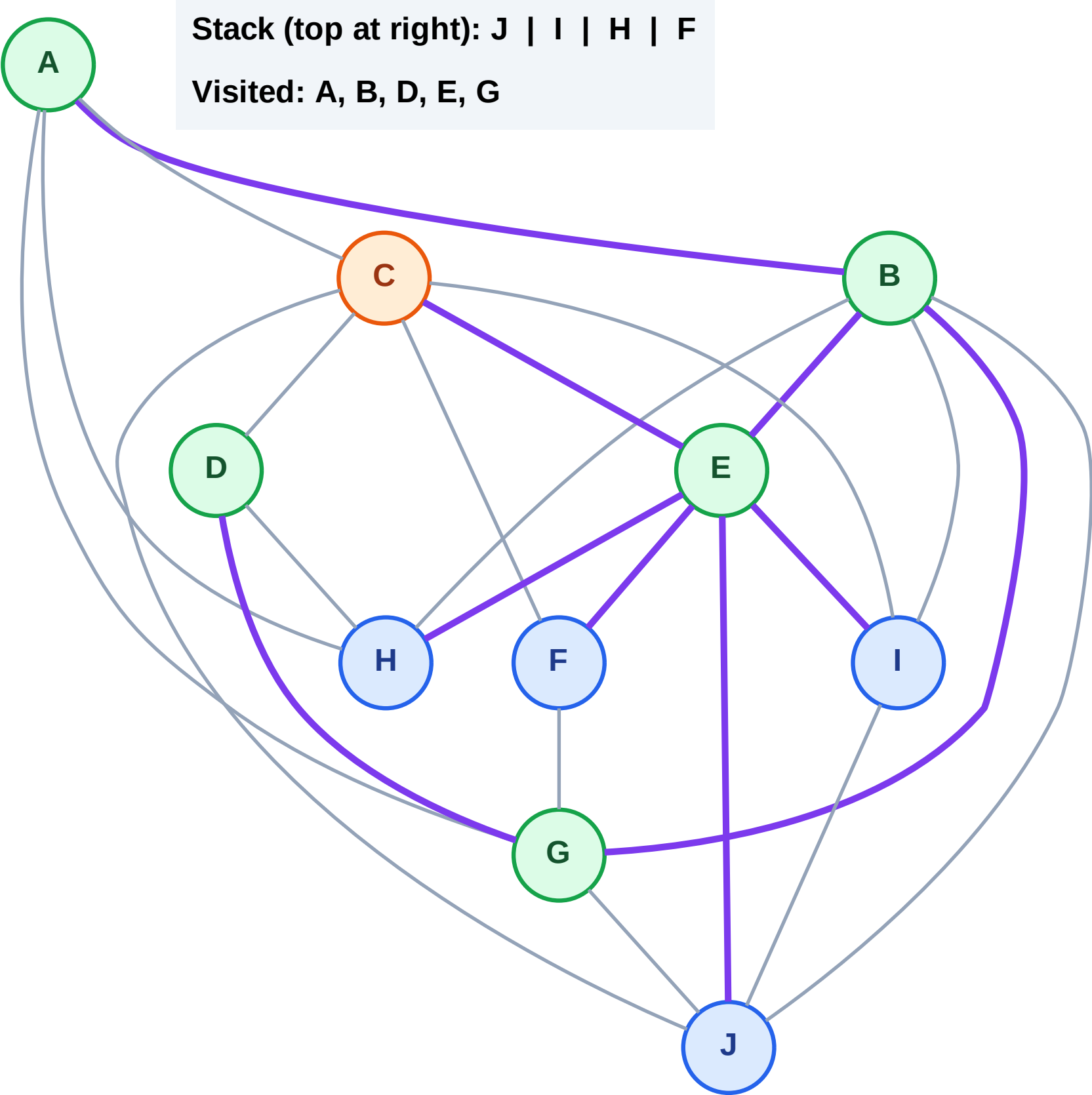
Legend

Complete

Processing

Discovered

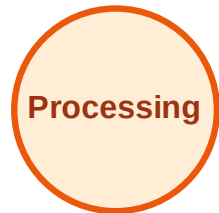
Unseen



# DFS Traversal

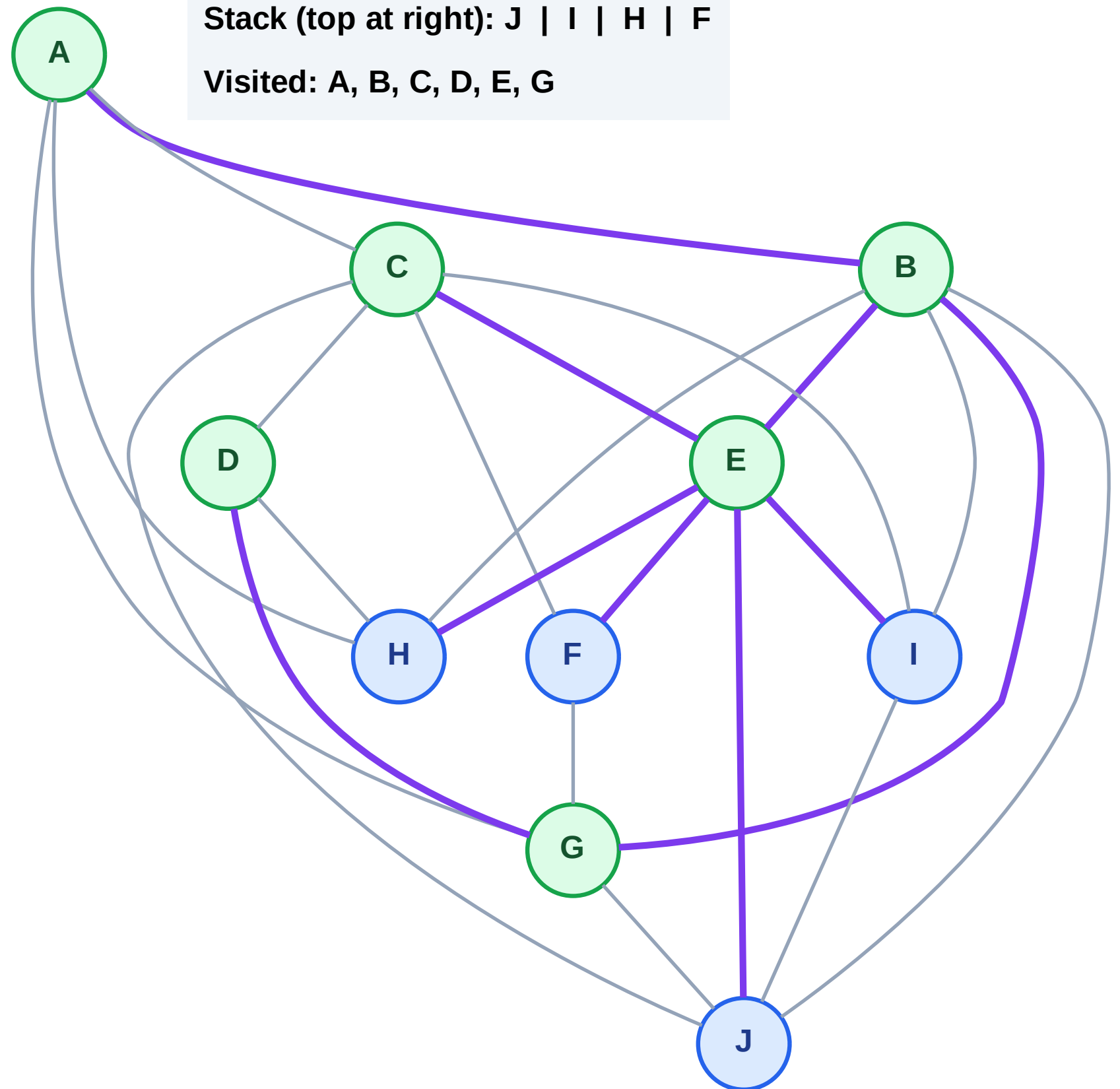
Step 13: No new neighbors from C

## Legend



Stack (top at right): J | I | H | F

Visited: A, B, C, D, E, G



# DFS Traversal

Step 14: Process node F

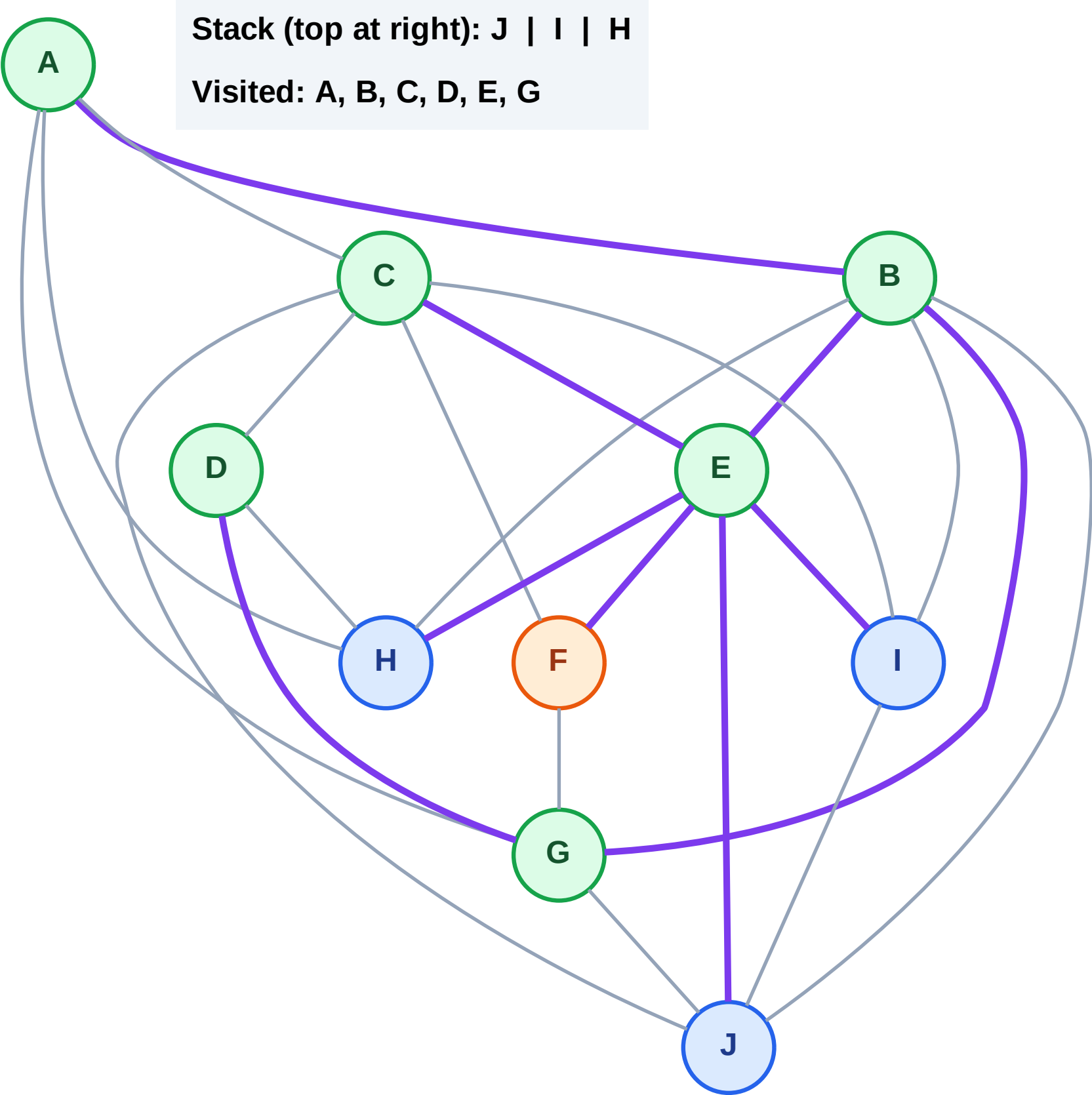
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 15: No new neighbors from F

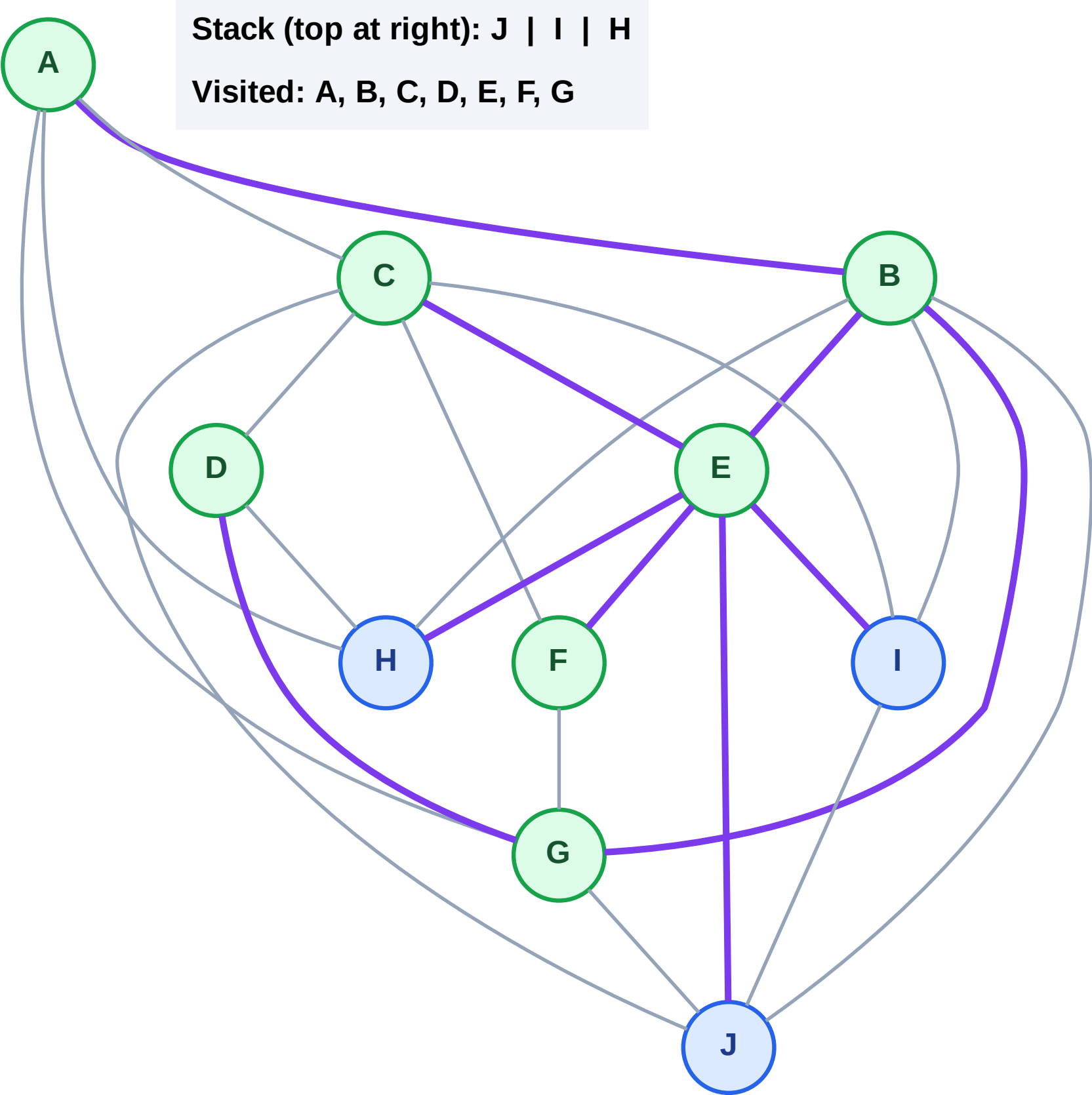
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 16: Process node H

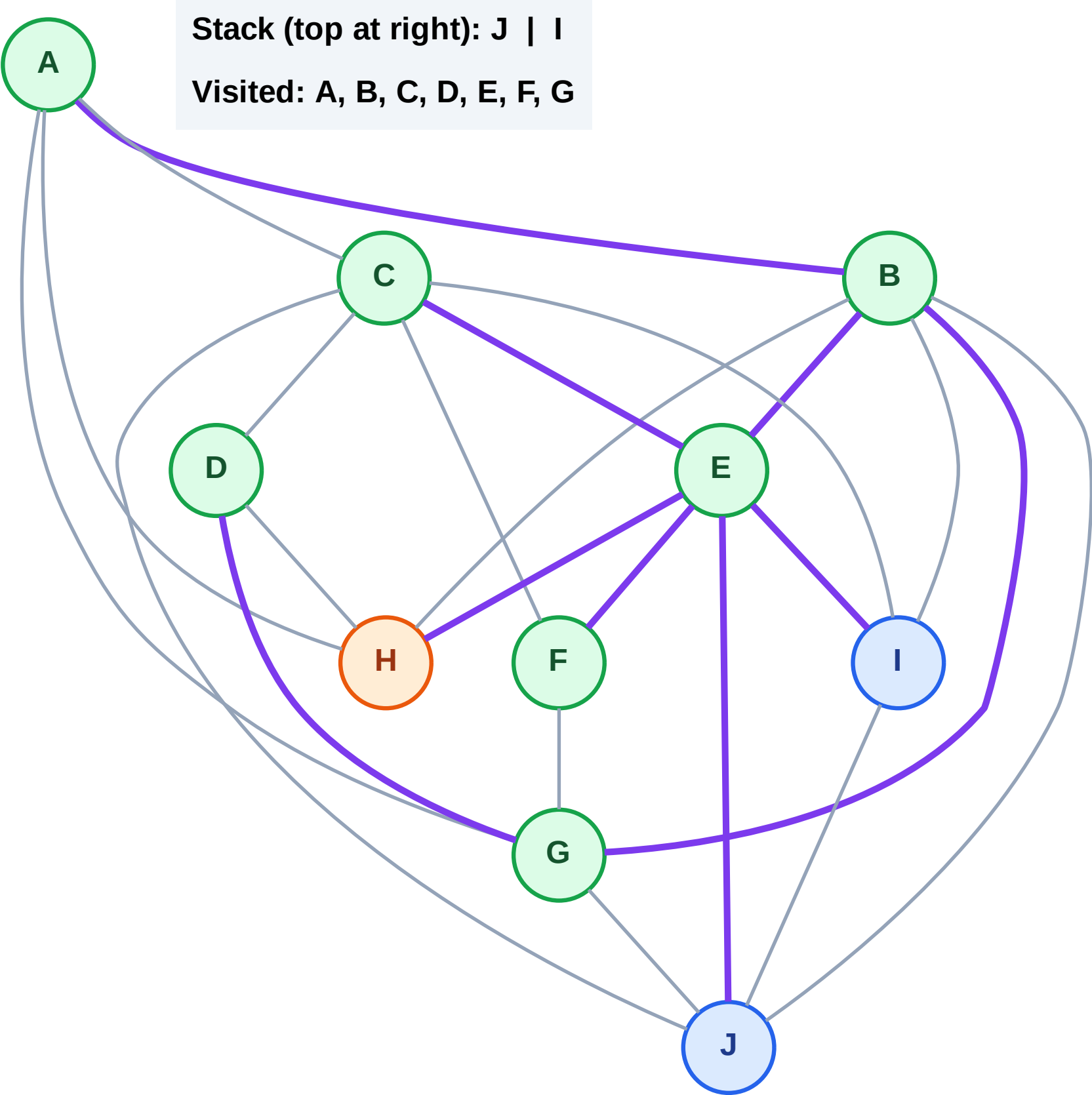
Legend

Complete

Processing

Discovered

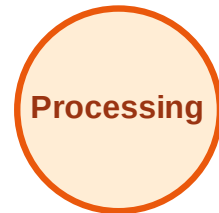
Unseen



# DFS Traversal

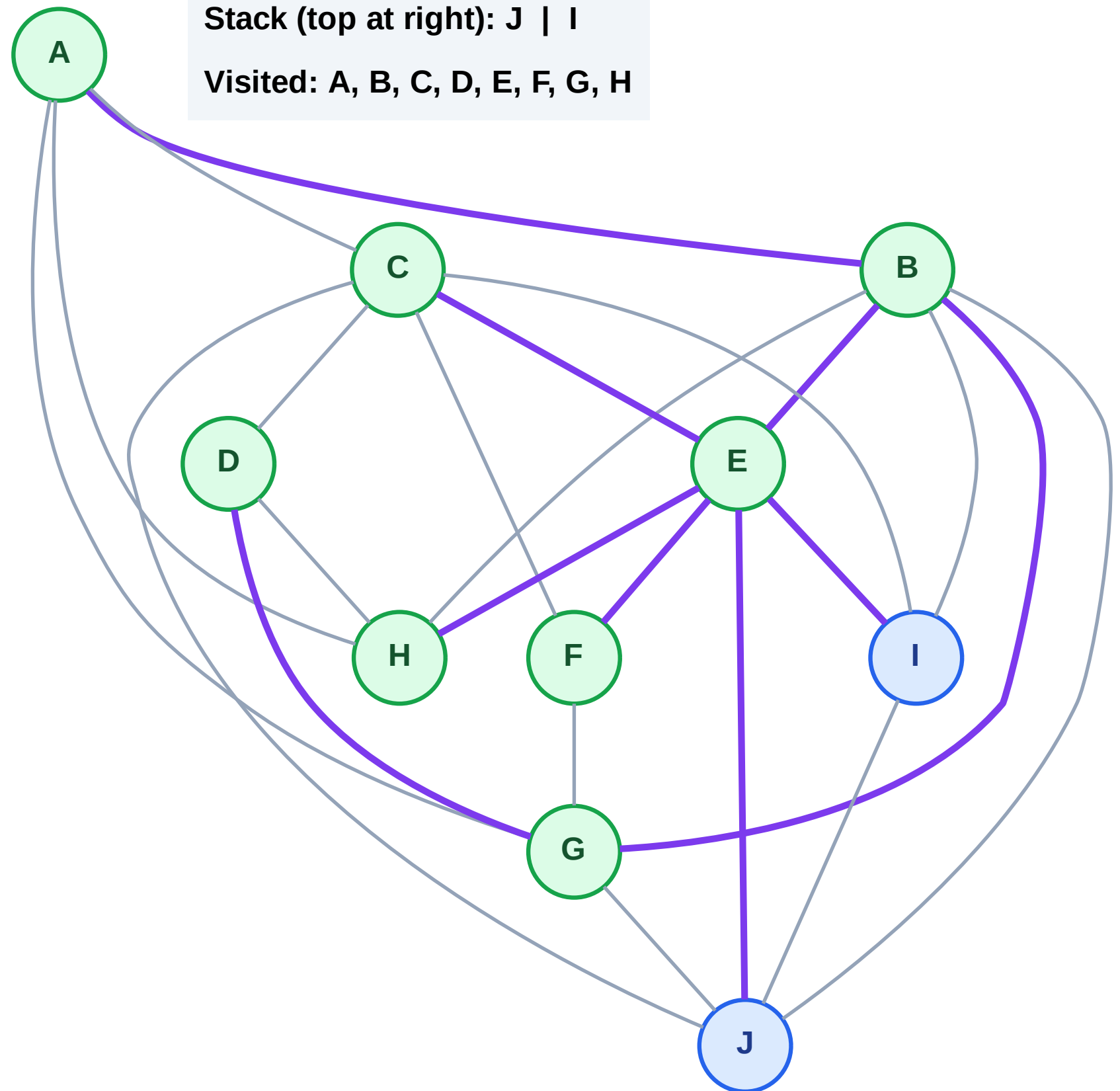
Step 17: No new neighbors from H

## Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



# DFS Traversal

Step 18: Process node I

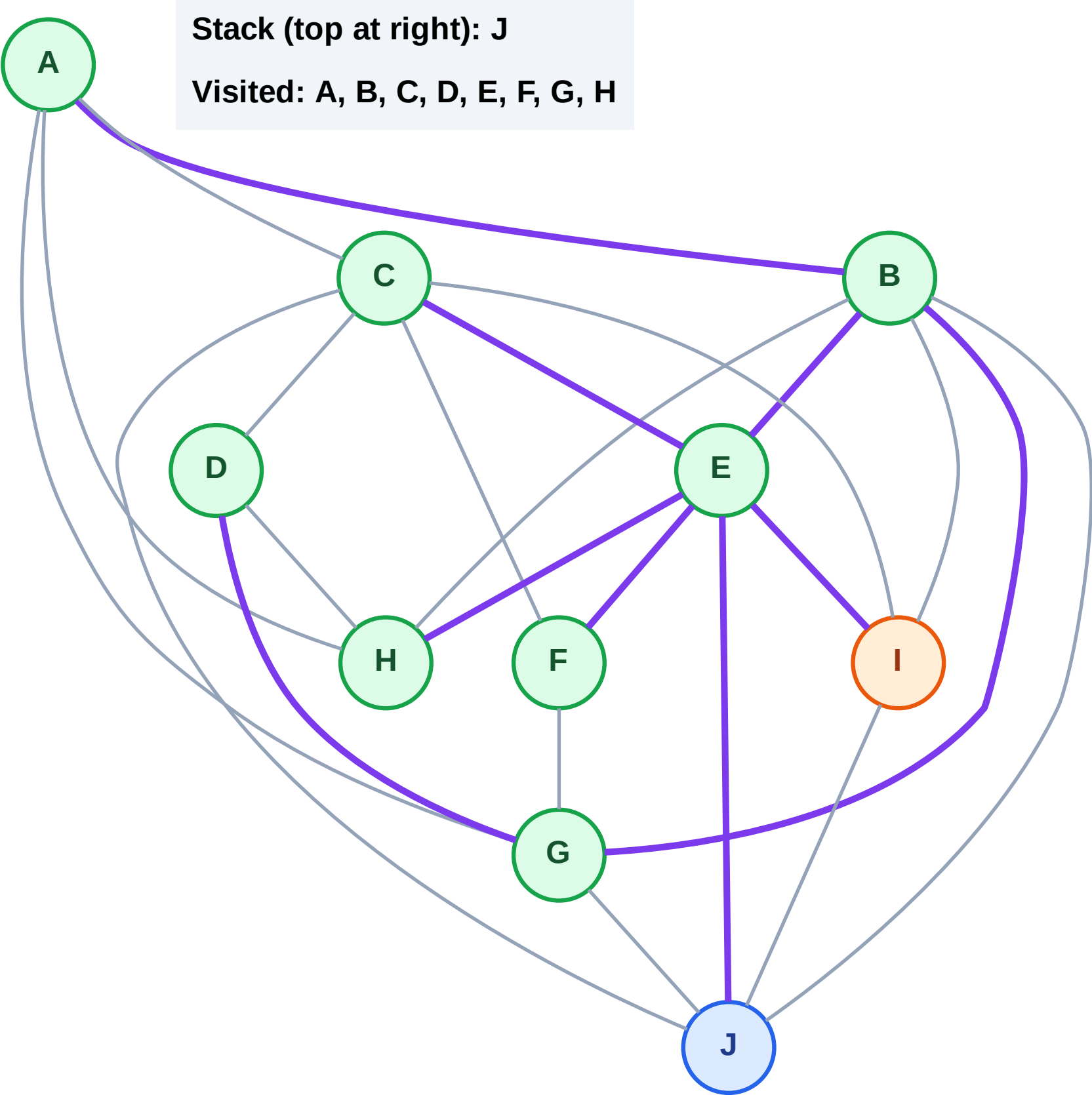
Legend

Complete

Processing

Discovered

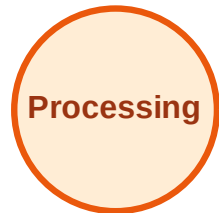
Unseen



# DFS Traversal

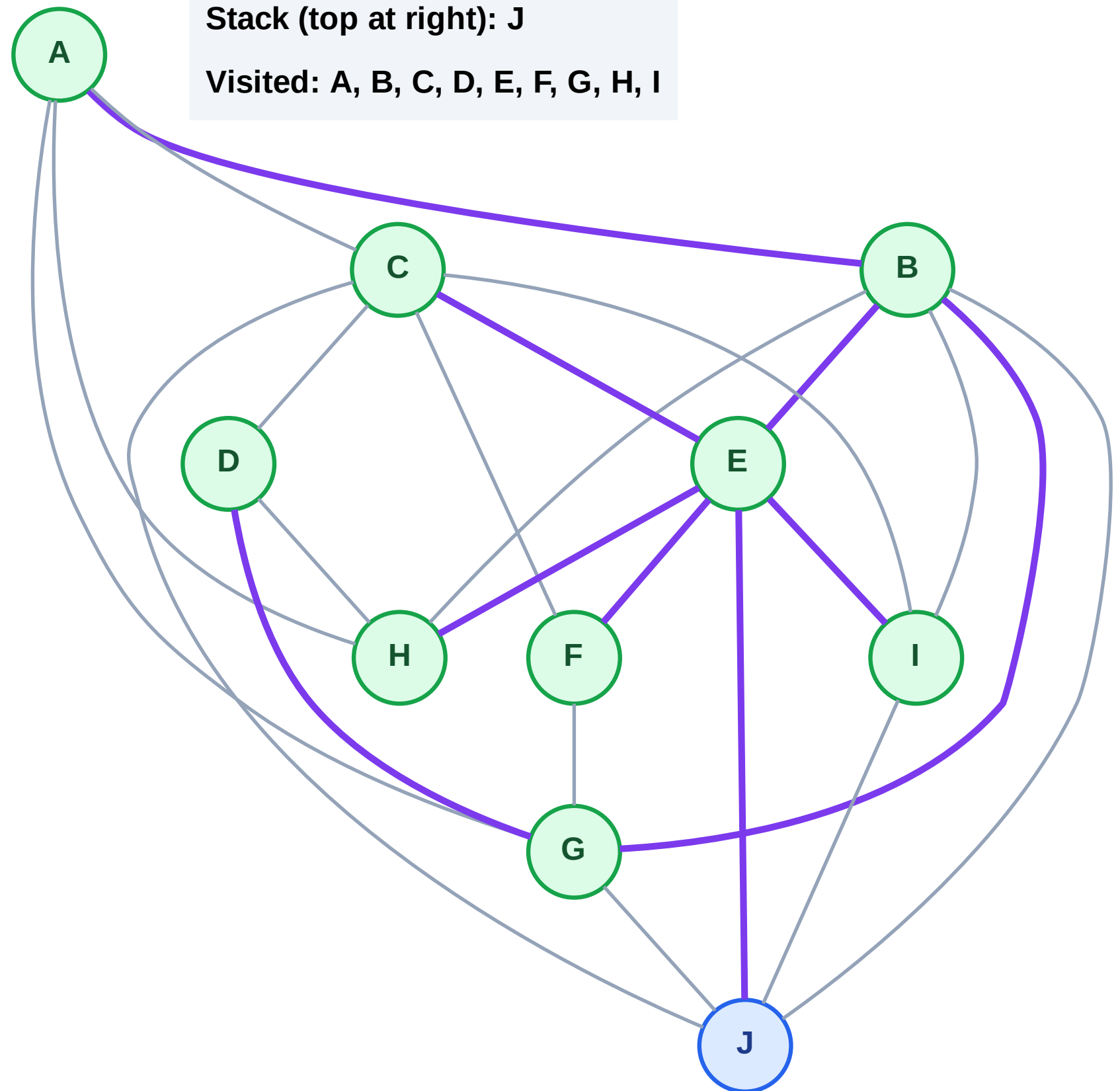
Step 19: No new neighbors from I

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 20: Process node J

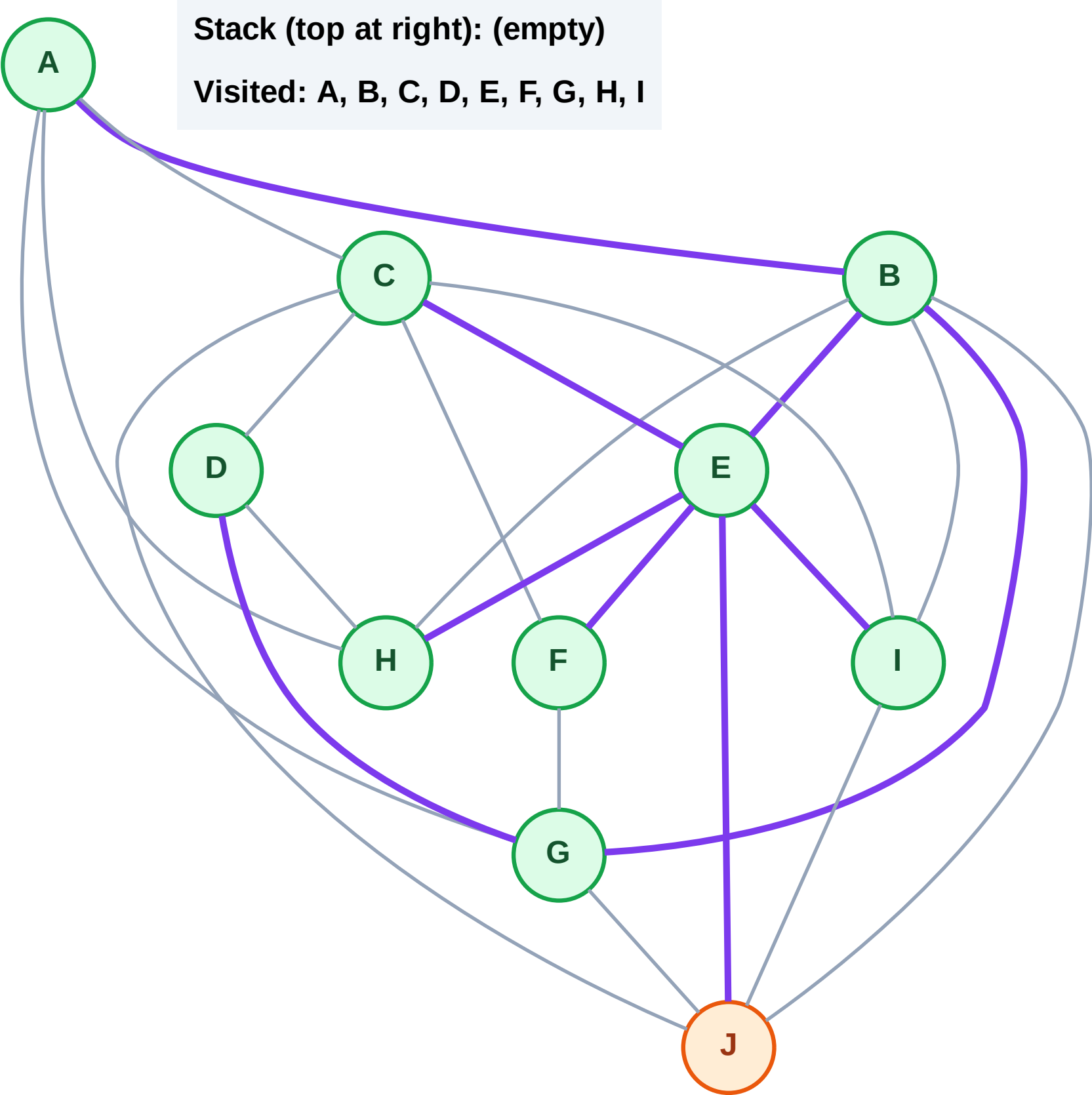
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

Processing

Discovered

Unseen

